

Balancing Costs and Utilities in Future Networks via Market Equilibrium with Externalities

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Abstract—Today, wireless networks are shifting towards systems that also involve computing resources, distributed across edge and cloud facilities. As such, radio and computing aspects shall be balanced continuously, to maximize the utilities of Service Providers (SPs), users quality of experience and fairness, while guaranteeing energy and carbon footprint constraints among others. In this paper, we tackle the problem of communication and compute resource allocation under energy constraints, with multiple SPs competing to get their preferred resource bundle by spending a fictitious currency budget. We model the system as a Fisher market (FM), incorporating energy use and carbon output as market externalities. Building on this framework, we develop a low-complexity, market-equilibrium (ME) based solution that ensures high utility, meets energy constraints, and promotes fairness among providers. To make the proposed resource allocation scheme practically viable and scalable, we design an alternating direction method of multipliers (ADMM) based equilibrium learning algorithm that enables SPs to reach the ME in a decentralized fashion. Finally, we run numerical simulations to validate the effectiveness of the proposed allocation mechanism, also for a practical use case of edge image classification, as well as the convergence rates of the distributed algorithm when scaling the number of players.

Index Terms—Green networking, Fisher market, market equilibrium, Pigouvian pricing, convex optimization, Nash welfare.

I. INTRODUCTION

Today, the Information and Communication Technology (ICT) sector represents around 2-4% of global CO₂ emissions. This includes devices, wireless network segments (i.e., Radio Access Networks - RANs), core and data centers, from production to operations. However, projections suggest that this global footprint share could increase in the near future [1]. This is mainly due to emerging new services that require not only communication resources, but also computing to be pervasive at all network layers, to process the myriad of data that are constantly generated by sensors for a plethora of applications also involving Artificial Intelligence (AI). As a result, unlike traditional mobile services, where radio resources are the primary resource, modern network services demand heterogeneous resources, including radio access capacity (or, communication resources), edge storage memory, and computational resources. One way to reduce the overall cost is to design and optimize 6G by taking into consideration Radio Access Network (RAN) segments [2] [3] and computing resources in a holistic fashion. To achieve this vision, legacy

energy efficiency metrics (e.g., in terms of energy per bit) fall short due to the potential rebound effects [4], and the described prominent role of computing. As an example, the authors of [5] indicate that a 5G network, despite its enhanced energy efficiency in J/bit, due to its larger bandwidth and better spatial multiplexing capabilities, could typically consume more than 4G, with a similar coverage area. Clearly, only dedicated policies (internal or external) such as budgeting resources (e.g., in terms of energy and/or CO₂) remain a valid way to limit the ICT footprint, to best balance it with the benefits. For instance, carbon quotas imply that future cumulative CO₂ must meet a given limit, thus they are a finite common global resource that must necessarily be shared among countries, whether through prior agreement or as an emergent property of individually determined national efforts [6]. In this work, we propose an approach, based on externalities, to introduce such resource budgeting. In particular, we focus on the challenging problem of competitive environments, in which a set of actors (e.g., Service Providers - SPs) aim to obtain a set of resources from the network, but need to allocate a limited budget that generates energy/CO₂ emissions costs. We follow an analytical approach that allows us to make claims on the convergence of a competitive game, both in centralized and distributed fashion. This model based approach needs to be adapted on a case by case considering involved resources and their energy footprint, as well as energy mix considerations.

A. Related works.

While network energy efficiency has been extensively studied, the focus shall shift towards absolute energy consumption-aware networking and, beyond that, carbon-aware management based on energy mix [7]. The authors of [8] outline ongoing standardization efforts, identify gaps in current mobile networks, and present a preliminary architectural perspective for energy-efficient, user-aware 6G networks. The work in [9] explores the vision of carbon-neutral 6G networks, outlining key enablers such as AI-driven optimization, energy harvesting, and green network architecture design to reduce carbon emissions across the communication ecosystem. The authors of [10] propose the concept of Decarbonization Service Agreements (DSAs) to incentivize sustainable behaviors among stakeholders by integrating energy and carbon-related performance indexes into SLAs, aiming to foster a holistic green economy across 6G.

Most existing work on energy-aware networking and carbon-neutral 6G systems primarily addresses the problem from a network-centric optimization perspective. These studies typically focus on network design, resource allocation

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architectures, and scheduling strategies that aim to minimize operational energy consumption while maintaining predefined bounds on key performance indicators (KPIs) such as throughput and delay. Typically, this is studied through the *Network Utility Maximization (NUM)* framework, with efficient decentralized algorithms developed under convexity assumptions [11], [12]. Strategic agent behavior has motivated *game-theoretic approaches* and incentive-compatible mechanisms such as *VCG* and *VCG–Kelly* [13]–[15]. However, standard NUM formulations may lead to unfair allocations, prompting the study of fairness notions including *proportional fairness*, *max–min fairness*, α -*fairness*, and *Nash bargaining* [16], [17].

The authors of [18] examine multi-resource allocation for network slicing under several fairness notions, including Ordered Weighted Averaging (OWA), Weighted Proportional Fairness (WPF), DRF, and the mood value rule, ultimately advocating OWA as a suitable fairness metric. More recently, Meskar et al. [19] introduced a multi-resource allocation scheme for mobile edge computing based on the Kalai–Smorodinsky bargaining solution. In addition, a growing body of literature continues to explore alternative fairness principles for resource allocation in modern networked systems, including the studies in [20]–[24]. These approaches typically require explicit knowledge of agents’ utility functions. In contrast, *market-based mechanisms*, notably *Competitive Equilibrium from Equal Incomes (CEEI)* or *Fisher markets*, embed fairness through budget allocation and decentralized decision-making, and have gained increasing attention for resource management in next-generation networks including cloud, fog computing, network slicing, and federated learning [25]–[30]. While prior work has primarily focused on allocation efficiency, in this paper we extend the FM framework by incorporating externalities, thereby enabling resource allocation to account for costs such as energy consumption. Cooperative resource allocation mechanisms—such as Nash bargaining and social welfare maximization—explicitly require agents to optimize a joint or group objective, often necessitating coordination, information sharing, or centralized enforcement. In contrast, the FM model assumes that agents are endowed with fixed budgets that they may spend freely on available resources. Each agent behaves rationally and myopically, maximizing only its own utility, while prices are determined through the market-clearing principle. Importantly, budget expenditure is verifiable by the market operator, ensuring that the mechanism remains fully decentralized and non-strategic. Unlike cooperative schemes, where welfare maximization is imposed as a design objective, the Fisher equilibrium achieves desirable welfare and fairness properties as an emergent consequence of decentralized budget-constrained utility maximization and price adjustment. Thus, while cooperative mechanisms rely on explicitly coordinated behavior, the Fisher model yields welfare-efficient and fair allocation through a competitive equilibrium, in which agents are assumed to be selfish and non-cooperative.

In summary, we identify the following gaps in the current literature:

- Existing green networking and carbon-neutral 6G studies

mainly focus on network-centric optimization, ignoring explicit energy consumption constraints;

- Existing NUM and cooperative mechanisms, such as Nash bargaining and social welfare maximization, require centralized coordination, knowledge of service providers’ utility functions, or cooperative behavior, limiting their applicability in decentralized systems with selfish service providers.
- Fisher market frameworks enable fair and decentralized resource allocation without requiring knowledge of participants’ utility functions. However, traditional FM models only balance supply and demand, without accounting for resource provisioning externalities.

B. Our contribution

We aim to fill these gaps, proposing an analytical framework to incorporate *environmental externalities*¹. In particular, we build on the Fisher market model, a special case of the Arrow-Debreu market [31], where buyers (agents) possess a single commodity used as currency or a fixed budget to acquire different resources (such as bandwidth, CPU, RAM, and storage). Whereas, market operators set prices to ensure resource capacity constraints are met. Eisenberg and Gale [32], later extended by Jain et al. [33], show that when buyers’ utilities are continuous, concave, and homogeneous of degree one (CCH)², market equilibrium can be determined via a convex optimization problem.

We extend this model by incorporating *externalities*, such as the environmental impact of energy consumption, into the market dynamics and demonstrate that equilibrium can still be computed via convex optimization. Our approach aligns with the principles of Pigouvian pricing [34], where we introduce taxes alongside prices that internalize external costs within the resource allocation mechanism, ensuring that both capacity constraints and external costs are satisfied, leading to a more sustainable and efficient allocation of resources.

Here below, we list the key contributions of this paper.

- 1) We consider a joint communication and compute resource allocation problem under energy (or, carbon) footprint constraints with multiple SPs competing to get their preferred resource bundle.
- 2) We introduce a novel notion of externalities into the classical FM framework, formulate the resource allocation problem as a Fisher market, and propose a ME-based solution.
- 3) We demonstrate that the ME can be obtained by solving a centralized optimization program and further propose an algorithm to efficiently compute the equilibrium in a decentralized manner.

¹An *externality* is a cost or benefit arising from actions of one market participant but affects the welfare of other agents or society without being reflected in market prices. In our setting, the energy consumption of resource usage generates environmental costs, such as CO₂ emissions, which are borne by the all the agents

²A function $f(x_1, x_2, \dots, x_n)$ is homogeneous of degree k if for any $t > 0$: $f(tx_1, tx_2, \dots, tx_n) = t^k f(x_1, x_2, \dots, x_n)$.

The remainder of the paper is organized as follows: Section II introduces the system model. In Section III, we present a Fisher Market with externalities and derive its equilibrium via an optimization-based solution. In Section IV, we cast the resource allocation problem as a Fisher market. In Section V, we provide an ADMM-based decentralized approach to compute the equilibrium of the formulated market. Section VI presents a performance evaluation of the proposed allocation scheme with extensive numerical simulations, also in the case of a practical scenario in which AI-based classification is offered as a service. Finally, in Section VII, we conclude the paper by summarizing the results and future work. A notation summary can be found in Table I.

II. SYSTEM MODEL

We consider a communication marketplace where a set of service providers, denoted by $\mathcal{S} = \{1, \dots, S\}$, acquire different types of resources from an infrastructure provider (InP) to offer services to end users. These resource types, indexed by the set $\mathcal{R} = \{1, \dots, R\}$, include radio spectrum, CPU, and RAM. The InP operates a network spanning multiple geographical regions, such as urban, residential, and industrial zones, collectively represented by a set $\mathcal{L} = \{1, \dots, L\}$, as illustrated in Fig. 1. To support services, each location $l \in \mathcal{L}$ is equipped with essential infrastructure, including base stations or cells for wireless connectivity, mobile edge computing (MEC) facilities for localized computation, and cloud computing resources shared across multiple locations to handle computationally intensive tasks. Let $\mathcal{C}$ denote the set of cloud computing facilities that can be used.

A. Service utilities and costs

Let $x_{lc}^s[r]$ represent the amount of resource r requested by SPs at location l from cloud facility c . For ease of notation, we define $c = 0$ to represent resources obtained from the MEC facility at the same location l . Let the parameter $d_{lc}^s[r]$ denote the base resource demand (or resource-consumption coefficient) of SPs for resource type r when serving users at location l from facility c . Specifically, $d_{lc}^s[r]$ represents the minimum amount of resource (r) required to support one unit of normalized service rate. Therefore, the ratio $x_{lc}^s[r]/d_{lc}^s[r]$ can be interpreted as the number of service units that can be supported by the allocated amount $x_{lc}^s[r]$ of resource r . A larger value of $d_{lc}^s[r]$ indicates lower resource efficiency, as more resources are required to provide the same service level.

Given an allocation $\mathbf{x}_{lc}^s = (x_{lc}^s[1], \dots, x_{lc}^s[R])$, the utility of the SPs is expressed as:

$$u_{lc}^s(\mathbf{x}_{lc}^s) = \min_r \left\{ \frac{x_{lc}^s[r]}{d_{lc}^s[r]} \right\} \quad \forall l \in \mathcal{L}, \forall c \in \mathcal{C} \quad (1)$$

The utility function in (1) captures a perfect complement relationship between resources. Since utility corresponds to the achieved service rate, $u_{lc}^s(\mathbf{x}_{lc}^s)$ represents the number of service units that can be supported by the allocated resource bundle. As an example, suppose a service has a base demand of 2 units of bandwidth (e.g., resource blocks) and 1 unit of CPU (e.g., 1 core). If it is allocated 4 units of bandwidth and 2

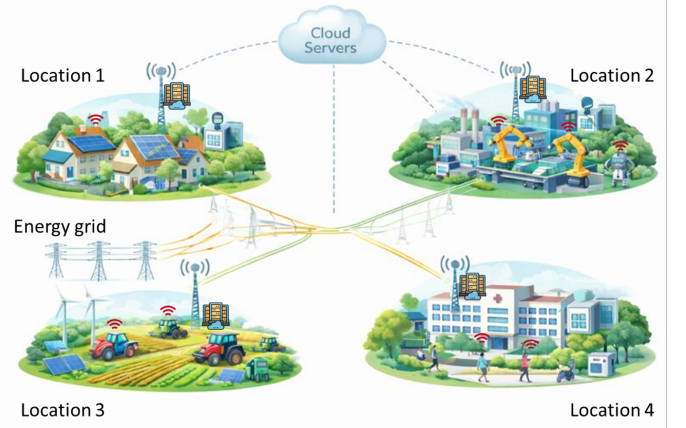


Fig. 1: Network spanning diverse geographical regions, such as residential, rural, hospital and industrial zones

units of CPU, its service rate is: $u_{lc}^s = \min \left\{ \frac{4}{2}, \frac{2}{1} \right\} = 2$. Now, if the bandwidth is increased to 6 units (while keeping CPU at 2 units), the utility remains the same. Thus, utility improves only when all allocated resources increase proportionally. This also motivates a joint approach in optimizing resources. Although we adopt a Leontief utility to capture the strict complementarity of computational resources (e.g., CPU and RAM), the proposed framework applies to the broader class of CCH utilities. For instance, Cobb–Douglas or CES utilities (with $\sigma > -\infty$) can also be accommodated to model partial substitutability. The restriction to Leontief utilities is made for modeling fidelity and analytical clarity, not due to a limitation of the framework.

The base resource demand ($d_{lc}^s[1], [r], \dots, d_{lc}^s[r]$) varies based on the proximity of the cloud or MEC facility to the service area. For example, achieving a unit service rate from a distant cloud requires more resources than from MEC, depending on its actual distance from the location. In particular, when users at location l are served by a distant cloud facility, additional communication overhead, latency constraints, and resource inefficiencies may require a larger amount of computing and networking resources to achieve the same service rate compared with a nearby MEC facility. Consequently, $d_{lc}^s[r]$ may vary across facilities and locations, capturing the heterogeneous efficiency of service provisioning in the cloud–edge infrastructure. The total utility obtained by SPs to support its service at location l from multiple cloud and MEC facilities is given by:

$$U_l^s(\mathbf{x}_l^s) = \sum_c u_{lc}^s(\mathbf{x}_{lc}^s). \quad (2)$$

where $\mathbf{x}_l^s = (\mathbf{x}_{lc}^s)_{c=0}^{c=C}$ denote the bundle of resources allocated to SP s at location l , from different cloud facilities including the MEC server located at l (i.e., for $c = 0$). The overall utility achieved by SPs across all locations is:

$$U^s(\mathbf{x}^s) = \sum_l U_l^s(\mathbf{x}_l^s) \quad (3)$$

Using network resources contributes, directly or indirectly, to CO₂ emissions. Since most of these emissions are due to

TABLE I: Notation Summary

Symbol	Description
$\mathcal{S}$	Set of service providers, $\{1, \dots, S\}$
$\mathcal{R}$	Set of resource types, $\{1, \dots, R\}$
$\mathcal{L}$	Set of geographical locations, $\{1, \dots, L\}$
$\mathcal{C}$	Set of cloud computing facilities
$x_{lc}^s[r]$	Amount of resource r requested by service s at location l from cloud facility c
$\mathbf{x}_{lc}^s$	Vector of resources r requested by s at l from c
$\mathbf{x}_l^s$	Bundle of resources allocated to s at location l from all facilities
$\mathbf{x}^s$	Total allocation vector for s across all locations and facilities
$d_{lc}^s[r]$	Base resource demand: minimum amount of resource r for service s at l, c to achieve unit utility
$u_{lc}^s(\mathbf{x}_{lc}^s)$	Utility of s at location l from facility c , defined as $\min_r \frac{x_{lc}^s[r]}{d_{lc}^s[r]}$
$U_l^s(\mathbf{x}_l^s)$	Total utility of s at location l , sum of utilities from all facilities c
$U^s(\mathbf{x}^s)$	Overall utility of s across all locations
$\hat{x}_l^s[r]$	Resource r requested by s from MEC at location l , i.e., x_{r10}^s
$\tilde{x}_c^s[r]$	Total resource r requested by s from cloud facility c , summed over all l
$\hat{\mathbf{x}}^s$	MEC resource vector for service s aggregated over r, l
$\tilde{\mathbf{x}}^s$	Cloud resource vector for service s aggregated over r, c
$\hat{\mathbf{x}}_{\text{avg}}^{(t)}, \tilde{\mathbf{x}}_{\text{avg}}^{(t)}$	Average MEC and cloud resource usage over all services at iteration t
$\hat{\mathbf{y}}^{(t)}, \tilde{\mathbf{y}}^{(t)}$	Auxiliary consensus variables for MEC and cloud at iteration t
$\hat{\mathbf{y}}_{\text{avg}}^{(t)}, \tilde{\mathbf{y}}_{\text{avg}}^{(t)}$	Average auxiliary consensus variables at iteration t
$\hat{\mathbf{p}}^{(t)}, \tilde{\mathbf{p}}^{(t)}$	Dual variables (Lagrange multipliers) for MEC and cloud at iteration t
B^s	Weight or budget of service provider s
ρ	ADMM penalty parameter
N	Number of service providers (sometimes $N = S$)
$\hat{e}_l(\hat{\mathbf{x}}_l)$	Energy consumption at location l as function of local resource usage
$\tilde{e}_c(\tilde{\mathbf{x}}_c)$	Energy consumption at cloud facility c as function of resource usage
E_l	Local energy consumption limit at location l
E_g	Global energy consumption limit over all locations and cloud facilities
$X_l[r]$	Capacity (physical limit) of resource r at location l

electricity use, we focus on the energy consumption associated with resource utilization to account for the carbon footprint of the overall communication and compute networks. Let $\hat{x}_l[r]$ represent the total demand for resource r (including both radio and MEC resources) from its local facility at location l . Similarly, let $\tilde{x}_c[r] = \sum_s \sum_l x_{lc}^s[r]$ represent the total demand for resource r at the cloud facility c , summed over all services s and locations l . In the direction of environmental sustainability, we assume that the total energy consumption, considering the carbon intensity of electricity generation, is subject to regulatory constraints. These constraints ensure that energy usage does not exceed specified limits set by regulatory authorities. In this work, we impose two constraints on various positions. First, we assume that each location l has a restriction on the total energy consumption. This energy consumption arises from two primary factors: the utilization of resources in the MEC facility available at that location (denoted by $c = 0$) and the energy consumed due to the total usage of radio bandwidth at that location. The latter includes the bandwidth used for uploading tasks to different cloud facilities as well as to MEC facilities. Let $\hat{e}_l(\hat{\mathbf{x}}_l)$ represent the total energy

consumed at location l , and $\tilde{e}_c(\tilde{\mathbf{x}}_c)$ represent the total energy consumed at cloud facility c . We assume that each location l has a local energy consumption limit denoted by E_l , i.e.

$$\hat{e}_l(\hat{\mathbf{x}}_l) \leq E_l \quad \forall l \in \mathcal{L} \quad (4)$$

Such a budget may arise from regional constraints, local power-supply or cooling limitations, dedicated energy grids, carbon budgets, or operator-defined sustainability policies. For instance, an edge site or data center powered partly by a dedicated renewable-energy grid may be assigned a different energy limit than a site relying entirely on the public grid. Moreover, E_l must not be interpreted as a permanently fixed quantity. In dynamic settings, it can be updated periodically to reflect variations in energy availability, carbon intensity, traffic demand, or operational policies, and the proposed allocation mechanism can be reapplied at each decision period. In addition to these local limits, the network is also subject to a global constraint, which caps the total energy across all locations and cloud facilities:

$$\sum_l \hat{e}_l(\hat{\mathbf{x}}_l) + \sum_c \tilde{e}_c(\tilde{\mathbf{x}}_c) \leq E_g \quad (5)$$

Each location $l \in \mathcal{L}$ is characterized by limited physical resources, bounded by a capacity $X_l[r]$ for each resource type r . Edge resources are therefore explicitly constrained. Although cloud resources are typically provisioned at a much larger scale, we incorporate capacity limits $X_c[r]$ to account for possible saturation under high service demand. Accordingly, the resource allocation must satisfy the following capacity constraints:

$$\hat{x}_l[r] \leq X_l[r], \quad \forall r, \forall l \in \mathcal{L}, \quad (6)$$

$$\tilde{x}_c[r] \leq X_c[r], \quad \forall r, \forall c \in \mathcal{C}. \quad (7)$$

We consider that each SP acts selfishly, aiming to maximize its individual utility. Service providers do not necessarily account for the impact of their resource usage on the collective emissions caused by other SPs. This behavior may lead to violations of the regulatory limits. The objective of this work is to design a pricing mechanism for resource usage that penalizes SPs, in order to maintain CO₂ emissions below the prescribed thresholds. However, when applying the mechanism, our goal is to harmonize the needs of all service providers and ensure a fair allocation. Toward this goal, we propose a Fisher Market-based resource allocation scheme.

III. FISHER MARKET MODEL WITH EXTERNALITIES

In this section, we introduce the FM model with externalities, which plays a central role in our proposed resource allocation framework. While the classical FM has been extensively studied in the context of supply-demand interactions and resource availability, the integration of external factors such as energy consumption and carbon emissions into the

market mechanism represents a novel extension. We define a Fisher market with externalities as

$$\mathcal{M} := \left\langle \mathcal{N}, (\mathbf{z}^n \in \mathbb{R}^K)_{n \in \mathcal{N}}, (\phi^n)_{n \in \mathcal{N}}, \right. \\ \left. (e_i)_{i \in \mathcal{I}}, (B^n)_{n \in \mathcal{N}}, \boldsymbol{\pi} \in \mathbb{R}^K \right\rangle,$$

where:

- *Buyers (agents)*: $\mathcal{N} = \{1, 2, \dots, N\}$ is the set of buyers, each requesting shares of divisible resources.
- *Resources*: $\mathcal{K} = \{1, 2, \dots, K\}$ is the set of resources. Each resource $k \in \mathcal{K}$ may be subject to capacity limits or be virtually unlimited. For limited resources,

$$\sum_{n \in \mathcal{N}} z_k^n \leq C_k, \quad \forall k \in \mathcal{K}. \quad (8)$$

For example, in our case (8), edge and cloud facility resources have limited capacity.

- *Allocation vector*: For buyer $n \in \mathcal{N}$, $\mathbf{z}^n = (z_1^n, z_2^n, \dots, z_K^n) \in \mathbb{R}^K$ denotes the allocated quantities of each resource. The total allocation of resource k is

$$\bar{z}_k = \sum_{n \in \mathcal{N}} z_k^n,$$

and $\bar{\mathbf{z}} = (\bar{z}_1, \dots, \bar{z}_K)$.

- *Utility functions*: Each buyer n has a utility function $\phi^n : \mathbb{R}^K \rightarrow \mathbb{R}_+$, representing the satisfaction or benefit from the allocated resources $\mathbf{z}^n$.
- *Externalities*: A set $\mathcal{I}$ indexes global externality constraints, e.g., energy consumption or carbon emissions. Each e_i represents the limit E_i on a global cost function $c_i(\bar{\mathbf{z}})$, so that

$$c_i(\bar{\mathbf{z}}) \leq E_i, \quad \forall i \in \mathcal{I}. \quad (9)$$

These constraints represent costs that are shared among all buyers. In our setting, for example, we model electricity consumption as a cost driver due to its associated CO₂ emissions. As an example for our setting, constraint (4) imposes a limit on local energy consumption, while constraint (5) enforces a total energy budget across all locations and cloud facilities.

- *Budgets*: Each buyer n is endowed with a budget $B^n > 0$, normalized so that $\sum_{n \in \mathcal{N}} B^n = 1$.
- *Prices*: The price vector is $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$, where π_k is the price per unit of resource k .

Given the prices $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$ set by the market operator, the feasible demand set of the agent n is defined as the set of demands satisfying its budget:

$$\mathcal{Z}^n(\boldsymbol{\pi}) = \left\{ \mathbf{z}^n \mid \mathbf{z}^n \in \mathbb{R}^K, \sum_k z_k^n \pi_k = B^n \right\} \quad (10)$$

We assume that agents exhibit rational self-interested behavior, with each agent seeking to maximize their individual utility. Therefore, the decision problem for each agent n is given as

$$P_n : \begin{aligned} & \underset{\mathbf{z}^n \geq 0}{\text{maximize}} && \phi^n(\mathbf{z}^n) \\ & \text{subject to} && \mathbf{z}^n \in \mathcal{Z}^n(\boldsymbol{\pi}), \end{aligned} \quad (11)$$

The goal of the market is to find an equilibrium between the individual objectives in (11), prices in such a way that each agent acts selfishly by requesting resources. The total demand for resources shall then satisfy both the energy constraints and the resource capacity limits.

Definition 1: A market equilibrium (ME) or competitive equilibrium (CE) in the market $\mathcal{M}$ is a pair $(\boldsymbol{\pi}^*, \mathbf{z}^*)$ of prices and allocation. At this equilibrium, the total energy constraints and resource capacities are met, and agents receive their most preferred resource bundles. Formally, $(\boldsymbol{\pi}^*, \mathbf{z}^*)$ constitutes a ME if the following two conditions hold:

C1 Given the resource price vector, every agent n spends its budget so as to receive resource bundle $\mathbf{z}^{n*}$ that maximizes its utility:

$$\mathbf{z}^{n*} \in \arg \max_{\mathbf{z}^n} \{ \phi^n(\mathbf{z}^n) \mid \mathbf{z}^n \in \mathcal{Z}^n(\boldsymbol{\pi}^*) \} \quad \forall n \in \mathcal{N} \quad (\text{C1.1})$$

C2 Either the constraints (8)–(9) are satisfied, (i.e., the cost from resource usage respects the given limits and the total usage does not exceed capacity) and will be positively priced; otherwise, that resource has zero price, i.e., we have:

$$\pi_k^* = \gamma_k^* + \sum_{i \in \mathcal{I}} \lambda_{ik}^*, \quad \forall k \in \mathcal{K} \quad (\text{C2.1})$$

$$\lambda_{ik}^* = \nabla_k e_i(\bar{\mathbf{z}}^*) \lambda_i^*, \quad \forall k \in \mathcal{K} \quad (\text{C2.2})$$

$$\lambda_i^* (e_i(\bar{\mathbf{z}}^*) - E_i) = 0, \quad \forall i \in \mathcal{I} \quad (\text{C2.3})$$

$$\gamma_k^* \left(\sum_n z_k^{n*} - C_k \right) = 0, \quad \forall k \in \mathcal{K} \quad (\text{C2.4})$$

In the expression $\pi_k^* = \gamma_k^* + \sum_{i \in \mathcal{I}} \lambda_{ik}^*$, the term γ_k^* represents the actual price arising from capacity constraints, while λ_{ik}^* can be interpreted as an additional tax (Pigovian price), imposed due to various cost restrictions. Next, we show that the equilibrium solution to the market $\mathcal{M}$ can be computed by solving a convex optimization problem.

A. Equilibrium Solution

Theorem 1: If each agent's utility function within market $\mathcal{M}$ is continuous, concave, and homogeneous of degree one (CCH), and $e_i(\bar{\mathbf{z}})$ is convex and increasing function in $\bar{\mathbf{z}}$, $\forall i \in \mathcal{I}$, the market equilibrium can be computed by solving an optimization program (12) where the optimal allocation $\mathbf{z}^*$ and the corresponding prices $\boldsymbol{\pi}^*$ (dual variables associated with constraints of the following optimization program as in (C2.1)–(C2.4)) represent the ME:

$$\begin{aligned} & \underset{\mathbf{z} \geq 0}{\text{maximize}} && \sum_{n \in \mathcal{N}} B^n \log(\phi^n(\mathbf{z}^n)) \\ & \text{subject to} && \\ & (\lambda_i) && e_i(\bar{\mathbf{z}}) \leq E_i, \quad \forall i \in \mathcal{I} \\ & (\gamma_k) && \sum_{n \in \mathcal{N}} z_k^n \leq C_k, \quad k \in \mathcal{K}, \end{aligned} \quad (12)$$

where $\lambda_i, \forall i \in \mathcal{I}$ and $\gamma_k, \forall k \in \mathcal{K}$ denote the Lagrange multipliers associated with the constraints.

Proof 1: The detailed proof is provided in Appendix A. The key idea is to establish the equivalence between the decentralized buyers' optimization problems and the centralized social welfare maximization problem. For a given price vector

π , we first derive the KKT conditions of each buyer's best-response problem in (11). A pair $(\pi^*, \mathbf{z}^*)$ is a ME if and only if $\mathbf{z}_n^*$ is an optimal solution of (11) i.e., (C1.1) for every buyer $n \in \mathcal{N}$ under prices π^* , and the market-clearing conditions (C2.1)–(C2.4) hold. We then derive the KKT conditions of the centralized problem (12) and show that they are equivalent to the buyers' KKT conditions together with the market-clearing conditions, where the equilibrium prices correspond to the dual variables of the resource-capacity constraints. Since the objective function is concave, the feasible set is convex, and the KKT conditions are necessary and sufficient for optimality, this equivalence establishes the result.

B. Nash welfare and fairness

We proved that the ME for $\mathcal{M}$ can be determined by solving the optimization program in (12). Building on the work of [32], we assert that ME allocations maximize Nash welfare. This follows from the fact that optimizing the objective function in (12) is equivalent to maximizing the Nash welfare function.

$$\arg \max_{\mathbf{z} \in \mathcal{Z}} \prod_n \phi^n(\mathbf{z}^n)^{B^n} = \arg \max_{\mathbf{z} \in \mathcal{Z}} \sum_{n \in \mathcal{N}} B^n \log(\phi^n(\mathbf{z}^n)) \quad (13)$$

Further, considering the first order condition for convex optimization

$$\sum_n B^n \frac{\phi^n(\mathbf{z}^n) - \phi^n(\mathbf{z}^{n*})}{\phi^n(\mathbf{z}^{n*})} \leq 0 \quad \forall \mathbf{z}, \mathbf{z}^* \in \mathcal{X} \quad (14)$$

we observe that the equilibrium utility vector $(\phi^1(\mathbf{z}^{1*}), \dots, \phi^N(\mathbf{z}^{N*}))$ can be interpreted as proportionally fair. Specifically, when comparing this equilibrium vector to any other feasible utility vector $(\phi^1(\mathbf{z}^1), \dots, \phi^N(\mathbf{z}^N))$, the sum of the proportional changes in utilities is non-positive. This implies that any deviation from the equilibrium utility allocation leads to a net negative or zero aggregate proportional change across all users. Therefore, the proposed allocation scheme maximizes Nash welfare or achieves the proportional fairness criteria while distributing the resources among the agents.

C. Examples of utility

One of the examples of a general class of CCH utility functions is the CES (constant elasticity of substitution) utility function, which is represented as $\phi_n(\mathbf{z}^n) = (\sum_k v_k^n (z_k^n)^\sigma)^{1/\sigma}$, where σ defines the function's elasticity. The parameter σ can range from $(-\infty, 0) \cup (0, 1]$. When $\sigma = 1$, the utility is linear, representing perfect substitutes. As $\sigma \rightarrow 0$, the utility function becomes Cobb-Douglas, which has a multiplicative form $\phi^n(\mathbf{z}^n) = \prod_k (z_k^n)^{v_k^n}$. As $\sigma \rightarrow -\infty$, it becomes Leontief $\phi^n(\mathbf{z}^n) = \min_k \{ \frac{z_k^n}{v_k^n} \}$, representing perfect complements. Linear valuation signifies that resources are perfect substitutes, whereas Leontief indicates perfect complements. The CES function interpolates between perfect substitutes and complements on the basis of the value σ . Moreover, applying the CES form to an existing CES utility function, such as $(\sum_k v_k^n (\phi^n(\mathbf{z}^n))^\sigma)^{1/\sigma}$, also results in a

CCH function, allowing for a broad range of utility representations. For Leontief utilities, the derivative is $\nabla_k^n \phi^n(\mathbf{z}^n) = \frac{1}{v_k^n}$ if k is binding constraint otherwise its value is 0, multiplying by z_k^n and summing, $\sum_{k=1}^K z_k^n \nabla_k^n \phi^n(\mathbf{z}^n) = z_k^n \frac{1}{v_k^n} = \phi^n(\mathbf{z}^n)$. Thus, Euler's theorem holds for the Leontief utility function.

The CES utility class encompasses a broad spectrum of resource interaction models, making the proposed framework applicable to a wide range of resource allocation problems. At one extreme, Leontief utilities model services that require resources in fixed proportions, such as virtual network functions or cloud-native applications whose performance is limited by the most constrained resource. At the other extreme, linear utilities represent scenarios where different resources can be substituted almost freely. Between these two cases, CES utilities capture varying degrees of substitutability and complementarity through the elasticity parameter (σ), while Cobb–Douglas utilities model situations where multiple resources jointly contribute to performance with diminishing marginal returns. This flexibility allows the framework to represent diverse service requirements encountered in future communication and computing systems. For example, latency-sensitive applications may exhibit strong complementarity between radio, computing, and memory resources and can be accurately modeled using Leontief-type utilities. In contrast, elastic workloads, content delivery services, or AI inference tasks may tolerate trade-offs among different resource types and are better represented by Cobb–Douglas or CES utilities. The CES class also supports hierarchical resource allocation through composition. In particular, the composition of CES utilities remains a CES utility, allowing resources allocated to a service to be further distributed among users or functions while preserving the utility structure. Moreover, the CES elasticity parameter can be used to control the fairness-efficiency trade-off, further broadening the applicability of the proposed framework [28] [20].

IV. MODELING RESOURCE ALLOCATION AS FM MODEL

We begin by formulating the resource allocation problem within the framework of the Fisher Market (FM) model, where service providers act as buyers, i.e., $\mathcal{N} = \mathcal{S}$. Each service provider $s \in \mathcal{S}$ is assigned a budget B^s , representing an artificial currency or *carbon credits*. These budgets are determined by a regulator and may reflect factors such as the provider's priority level or their efforts in reducing carbon emissions—e.g., by integrating renewable energy sources into their operations. With this budget, each SP can acquire resources distributed across various locations and cloud facilities, denoted collectively by $\mathcal{K} = \{(r, l, c) \in \mathcal{R} \times \mathcal{L} \times \mathcal{C}\}$. These resources generate utility for the service providers, with each buyer's utility function denoted by $\phi^n = U^s$. A central market operator is responsible for setting the prices of these resources. Service providers must pay these prices—potentially inclusive of externality-based taxes—to access the desired resources. Let $p_{lc}[r]$ denote the unit price of resource $r \in \mathcal{R}$ at location l in cloud facility c . The corresponding price vector for cloud facility c at location l is defined as $\mathbf{p}_{lc} = (p_{lc}[r])_{r \in \mathcal{R}}$, and the collection of all price vectors across locations and cloud

facilities is denoted by $\mathbf{p} = (\mathbf{p}_{lc})_{l \in \mathcal{L}, c \in \mathcal{C}}$. The market operator sets prices to maximize resource utilization while guiding the allocation toward socially or environmentally desirable outcomes, ensuring that resource usage and energy consumption stay within prescribed limits.

A. Service providers problem

We consider that the SPs are rational and given the announced prices, they allocate their budgets across different resources in different locations to procure the optimal bundle of heterogeneous resources required to support their services. Thus, the decision problem for each SP is defined as follows:

$$\begin{aligned} Q_s : \quad & \max_{\mathbf{x}^s \geq 0} \sum_{l \in \mathcal{L}} \sum_{c \in \mathcal{C} \cup \{0\}} u_{lc}^s(\mathbf{x}_{lc}^s) \\ \text{s.t.} \quad & \sum_{l \in \mathcal{L}} \sum_{c \in \mathcal{C} \cup \{0\}} \mathbf{p}_{lc}^\top \mathbf{x}_{lc}^s \leq B^s, \\ & \sum_{c \in \mathcal{C} \cup \{0\}} u_{lc}^s(\mathbf{x}_{lc}^s) \leq \hat{U}_l^s \quad \forall l \in \mathcal{L}. \end{aligned}$$

with $u_{lc}^s(\mathbf{x}_{lc}^s) = \min_r \left\{ \frac{x_{lc}^s[r]}{d_{lc}^s[r]} \right\}$, $\forall c \in \mathcal{C}, l \in \mathcal{L}$. In the above program, note that the SPs' decisions (*i.e.*, resource demand) depend on the prices set for the resources. However, given any prices, there is no guarantee that the total demand for each resource will remain within the resource capacity or that the energy consumption due to its usage will remain within the allowed limits.

Definition 2: A market equilibrium (ME) for the market $\mathcal{M}$ is defined as a pair $(\mathbf{p}^*, \mathbf{x}^*)$ of prices and allocation, where the market meets its total energy restrictions and service providers get their preferred resource bundle. Mathematically $(\mathbf{p}^*, \mathbf{x}^*)$ is a CE if the following two conditions are satisfied.

C1 Given the price vector, $\forall s \in \mathcal{S}$ spends its budget such that it receives resource bundle $\mathbf{x}^{s*}$ that maximizes its utility.

$$\mathbf{x}^{s*} \in \arg \max \left\{ U^s(\mathbf{x}^s) \mid \sum_l \sum_c \mathbf{p}_{lc}^\top \mathbf{x}_{lc}^s \leq B^s \right\}, \quad (\text{C1})$$

If the total energy consumption due to resources usage meets the capacity, it is positively priced; otherwise, the corresponding resource has zero price, *i.e.*, we have:

$$p_{rl}^* = \nu_{rc} + \lambda \nabla_{\tilde{x}_c[r]} \tilde{e}_c(\tilde{x}_c^*) \quad \forall r \in \mathcal{R}, \forall l \in \mathcal{L}, \forall c \in \mathcal{C} \setminus \{0\}, \quad (\text{C1.1})$$

$$p_{rl0}^* = \gamma_{rl} + (\lambda + \mu_l) \nabla_{\hat{x}_l[r]} \hat{e}_l(\hat{x}_l^*) \quad \forall r \in \mathcal{R}, \forall l \in \mathcal{L}, \quad (\text{C1.2})$$

$$\gamma_{rl}(\hat{x}_l^*[r] - X_l[r]) = 0 \quad \forall r \in \mathcal{R}, \forall l \in \mathcal{L}, \quad (\text{C1.3})$$

$$\nu_{rc}(\tilde{x}_c^*[r] - X_c[r]) = 0 \quad \forall r \in \mathcal{R}, \forall c \in \mathcal{C}, \quad (\text{C1.4})$$

$$\mu_l(\hat{e}_l(\mathbf{x}^*) - E_l) = 0 \quad \forall l \in \mathcal{L}, \quad (\text{C1.5})$$

$$\lambda(\sum_l \hat{e}_l(\hat{x}_l^*) + \sum_c \tilde{e}_c(\tilde{x}_c^*) - E_g) = 0 \quad (\text{C1.6})$$

In the above conditions, γ_{rl} and ν_{rc} represent the actual price per unit of resource r at edge location l and cloud c , respectively. μ_l denotes an additional tax applied at location l due to local energy consumption limits, and λ is a global tax imposed to ensure that overall resource usage remains below a specified threshold. In the next section, we demonstrate that the ME solution to the formulated market can be computed by solving the convex optimization program.

B. Solution

Proposition 1: Consider a market where each agent's utility is defined as in (1)–(3). Suppose that the functions $\hat{e}_l(\hat{x}_l)$ and $\tilde{e}_c(\tilde{x}_c)$ are convex and increasing in $\hat{x}_l$ and $\tilde{x}_c$, respectively, for all $l \in \mathcal{L}$ and $c \in \mathcal{C}$. Then, ME can be obtained by solving the optimization problem (15). Also, the optimal allocation $\mathbf{x}^*$ and the associated dual variables $\mathbf{p}^*$ (corresponding to constraints (C1.1)–(C1.6)) together constitute a ME.

$$\begin{aligned} & \text{maximize}_{\mathbf{x} \geq 0} \quad \sum_s B^s \log(U^s(\mathbf{x}^s)) \\ & \text{subject to} \quad \text{equations (1) – (3)} \\ & (\gamma_{rl}) \quad \hat{x}_l[r] - X_l[r] = 0, \forall r \in \mathcal{R}, \forall l \in \mathcal{L}, \\ & (\nu_{rc}) \quad \tilde{x}_c[r] - X_c[r] = 0, \forall r \in \mathcal{R}, \forall l \in \mathcal{C}, \\ & (\mu_l) \quad \hat{e}_l(\hat{x}_l) \leq E_l, \forall l \in \mathcal{L}, \\ & (\lambda) \quad \sum_l \hat{e}_l(\hat{x}_l) + \sum_c \tilde{e}_c(\tilde{x}_c) \leq E_g. \end{aligned} \quad (15)$$

Proof 2: As the utility function of each SP (agent) is concave homogeneous of degree one, and $\hat{e}_l(\hat{x}_l)$ and $\tilde{e}_c(\tilde{x}_c)$ are convex and increasing in $\hat{x}_l$ and $\tilde{x}_c$, the claim directly follows from Theorem 1.

We have shown that ME can be found by solving the optimization program in (15). This program's objective is equivalent to maximizing the Nash social welfare. As a result, the equilibrium allocation also maximizes the Nash welfare or ensures proportional fairness while distributing resources among the SPs.

V. DECENTRALIZED ADMM-BASED SOLUTION

In this section, we design a decentralized allocation algorithm based on the Alternating Direction Method of Multipliers (ADMM), enabling service providers (SPs), modeled as agents, to reach a ME without centralized coordination. To facilitate decentralized coordination, we introduce auxiliary variables and dual variables that enforce global consensus through local updates. Each agent s maintains local estimates of resource allocations, denoted by $\hat{\mathbf{x}}^{s(t)}$ for MEC (Multi-access Edge Computing) usage and $\tilde{\mathbf{x}}^{s(t)}$ for cloud energy usage at iteration t . The average of these local variables across all N agents is given by

$$\hat{\mathbf{x}}_{\text{avg}}^{(t)} = \frac{1}{S} \sum_{s'=1}^S \hat{\mathbf{x}}^{s'(t)}, \quad \tilde{\mathbf{x}}_{\text{avg}}^{(t)} = \frac{1}{S} \sum_{s'=1}^S \tilde{\mathbf{x}}^{s'(t)}.$$

To enforce agreement between agents, we introduce global auxiliary (consensus) variables $\hat{\mathbf{y}}^{(t)}$ and $\tilde{\mathbf{y}}^{(t)}$, representing shared MEC and cloud usage, whose averages are denoted by

$$\hat{\mathbf{y}}_{\text{avg}}^{(t)} = \frac{1}{S} \hat{\mathbf{y}}^{(t)}, \quad \tilde{\mathbf{y}}_{\text{avg}}^{(t)} = \frac{1}{S} \tilde{\mathbf{y}}^{(t)}.$$

The dual variables $\hat{\mathbf{p}}^{(t)}$ and $\tilde{\mathbf{p}}^{(t)}$ are associated with the coupling constraints $\hat{\mathbf{x}} = \hat{\mathbf{y}}$ and $\tilde{\mathbf{x}} = \tilde{\mathbf{y}}$, respectively. These are updated iteratively to drive local variables toward global consensus. Finally, the scalar parameter $\rho > 0$ denotes the ADMM penalty coefficient, which balances constraint enforcement

Algorithm 1 Semi-Decentralized ADMM for Resource Allocation with Adaptive Penalty

```

1: Initialize:  $\mathbf{x}^{s(0)}, \mathbf{p}^{(0)}$ , choose  $\rho^{(0)} > 0$ , parameters  $\mu > 1$ ,
    $\tau_{\text{inc}} > 1$ ,  $\tau_{\text{dec}} > 1$ ,  $\varepsilon_r > 0$ ,  $\varepsilon_s > 0$ 
2: for  $t = 0, 1, 2, \dots$  do
3:   Each agent  $s$  (in parallel):
4:     Receive  $\hat{\mathbf{x}}_{\text{avg}}^{(t)}, \tilde{\mathbf{x}}_{\text{avg}}^{(t)}, \hat{\mathbf{y}}_{\text{avg}}^{(t)}, \tilde{\mathbf{y}}_{\text{avg}}^{(t)}, \hat{\mathbf{p}}^{(t)}, \tilde{\mathbf{p}}^{(t)}, \rho^{(t)}$ 
5:     solve its local convex optimization problem (32)
6:     Send  $\hat{\mathbf{x}}^{s(t+1)}, \tilde{\mathbf{x}}^{s(t+1)}$  to operator
7:   Central operator:
8:     Compute averages:
9:      $\hat{\mathbf{x}}_{\text{avg}}^{(t+1)} = \frac{1}{N} \sum_s \hat{\mathbf{x}}^{s(t+1)}, \tilde{\mathbf{x}}_{\text{avg}}^{(t+1)} = \frac{1}{N} \sum_s \tilde{\mathbf{x}}^{s(t+1)}$ 
10:    Computes a projection onto the feasible set  $\mathcal{C}$  by (34)
11:    Compute averaged projection:
12:     $\hat{\mathbf{y}}_{\text{avg}}^{(t+1)} = \frac{1}{N} \hat{\mathbf{y}}^{(t+1)}, \tilde{\mathbf{y}}_{\text{avg}}^{(t+1)} = \frac{1}{N} \tilde{\mathbf{y}}^{(t+1)}$ 
13:    Dual updates:
14:     $\hat{\mathbf{p}}^{(t+1)} = \hat{\mathbf{p}}^{(t)} + \rho^{(t)} (\hat{\mathbf{x}}_{\text{avg}}^{(t+1)} - \hat{\mathbf{y}}_{\text{avg}}^{(t+1)})$ 
15:     $\tilde{\mathbf{p}}^{(t+1)} = \tilde{\mathbf{p}}^{(t)} + \rho^{(t)} (\tilde{\mathbf{x}}_{\text{avg}}^{(t+1)} - \tilde{\mathbf{y}}_{\text{avg}}^{(t+1)})$ 
16:    Compute primal residual:
17:     $\mathbf{r}^{(t+1)} = [\hat{\mathbf{x}}_{\text{avg}}^{(t+1)} - \tilde{\mathbf{y}}_{\text{avg}}^{(t+1)}, \tilde{\mathbf{x}}_{\text{avg}}^{(t+1)} - \hat{\mathbf{y}}_{\text{avg}}^{(t+1)}]$ 
18:    Compute dual residual:
19:     $\mathbf{s}^{(t+1)} = \rho^{(t)} [\hat{\mathbf{y}}_{\text{avg}}^{(t+1)} - \hat{\mathbf{y}}_{\text{avg}}^{(t)}, \tilde{\mathbf{y}}_{\text{avg}}^{(t+1)} - \tilde{\mathbf{y}}_{\text{avg}}^{(t)}]$ 
20:    Compute penalty:
21:     $\rho^{(t+1)} \leftarrow \rho^{(t+1)}$  by Algorithm 2 (see Appendix)
22:    Broadcast updated values
23:     $\hat{\mathbf{x}}_{\text{avg}}^{(t+1)}, \tilde{\mathbf{x}}_{\text{avg}}^{(t+1)}, \hat{\mathbf{y}}_{\text{avg}}^{(t+1)}, \tilde{\mathbf{y}}_{\text{avg}}^{(t+1)}, \hat{\mathbf{p}}^{(t+1)}, \tilde{\mathbf{p}}^{(t+1)}, \rho^{(t+1)}$ 
24:    to agents
25: end for

```

with objective minimization. Now we define global constraint set in terms of energy restriction and resource capacities $\mathcal{C}$ is

$$\mathcal{C} := \left\{ \begin{array}{l} (\hat{\mathbf{y}}, \tilde{\mathbf{y}}) : \hat{Y}_l[r] \leq X_l[r], \forall r \in \mathcal{R}, \forall l \in \mathcal{L}, \\ \tilde{Y}_c[r] \leq X_c[r], \forall r \in \mathcal{R}, \forall c \in \mathcal{C}, \\ \hat{e}_l(\hat{\mathbf{y}}_l) \leq E_l, \forall l \in \mathcal{L}, \\ \sum_l \hat{e}_l(\hat{\mathbf{y}}_l) + \sum_c \tilde{e}_c(\tilde{\mathbf{y}}_c) \leq E_g \end{array} \right\} \quad (16)$$

We rewrite the optimization problem (15) as

$$\begin{aligned} & \underset{\{\mathbf{x}^s\}}{\text{maximize}} && \sum_s B^s \log(U^s(\mathbf{x}^s)) \\ & \text{subject to} && \hat{\mathbf{x}}^s = \hat{\mathbf{y}}^s, \quad \tilde{\mathbf{x}}^s = \tilde{\mathbf{y}}^s, \quad \forall s \\ & && (\hat{\mathbf{y}}, \tilde{\mathbf{y}}) \in \mathcal{C} \end{aligned}$$

and consider the augmented Lagrangian:

$$\begin{aligned} \mathcal{L}_\rho(\{\mathbf{x}^s\}, \hat{\mathbf{y}}, \tilde{\mathbf{y}}, \hat{\mathbf{p}}, \tilde{\mathbf{p}}) := & \sum_s B^s \log \left(\sum_{l,c} u_{lc}^s(\mathbf{x}_{lc}^s) \right) \\ & + \sum_s \langle \hat{\mathbf{p}}^s, \hat{\mathbf{x}}^s - \hat{\mathbf{y}}^s \rangle + \sum_s \langle \tilde{\mathbf{p}}^s, \tilde{\mathbf{x}}^s - \tilde{\mathbf{y}}^s \rangle \\ & - \frac{\rho}{2} \sum_s \|\hat{\mathbf{x}}^s - \hat{\mathbf{y}}^s\|^2 - \frac{\rho}{2} \sum_s \|\tilde{\mathbf{x}}^s - \tilde{\mathbf{y}}^s\|^2 \\ & + \mathbb{I}_{\mathcal{C}}(\hat{\mathbf{y}}, \tilde{\mathbf{y}}) \end{aligned}$$

Based on the augmented Lagrangian we developed the ADMM based resource allocation scheme

In the decentralized ADMM framework the dual variables are updated collectively in the current presentation for notational simplicity, the algorithm naturally supports maintaining

TABLE II: Complexity Summary of the Proposed Algorithm

Operation	Complexity
Local optimization (per agent)	$O(K^3)$
Aggregate local updates	$O(SK^3)$
Central aggregation	$O(SI)$
Projection onto $\mathcal{C}$	$P(I)$
Dual + adaptive updates	$O(I)$
Communication per iteration	$O(SI)$
Memory complexity	$O(S(K+I))$
Per-iteration complexity	$O(SK^3 + SI + P(I))$
Overall complexity	$O\left(\frac{SK^3 + SI + P(I)}{\epsilon}\right)$

distinct dual variables for each class of constraint, such as resource capacity constraints and energy-related constraints. Under this interpretation, the multipliers γ_{rl} and ν_{rc} represent capacity prices associated with edge and cloud resources, respectively, whereas μ_l and λ correspond to local and global Pigouvian taxes induced by energy consumption limits. By organizing the dual variables according to constraint type, the Infrastructure Provider (InP) or a regulator can independently compute, monitor, and communicate the capacity prices and environmental tax signals.

A. Computational Complexity Analysis

Let S denote the number of service providers, $R = |\mathcal{R}|$ the number of resource types, $L = |\mathcal{L}|$ the number of locations, and $C = |\mathcal{C}|$ the number of cloud/MEC facilities. We define $K = RLC$ as the number of resource allocation variables associated with each service provider, and let I denote the dimension of the consensus (coupling) variables. Furthermore, let $P(I)$ represent the computational complexity of the projection onto the feasible set $\mathcal{C}$. Table II summarizes the computational, communication, and memory complexities of the proposed ADMM-based resource allocation algorithm. The dominant computational cost arises from solving the local convex optimization problem at each service provider, while the aggregation, projection, and dual-update steps incur lower-order complexity.

VI. NUMERICAL SIMULATIONS

For our numerical experiment, we first examine a scenario involving ten competing SPs operating across five distinct locations. Each SP offers a unique type of application, distinguished by its primary resource demand: (1) radio-intensive, (2) computing-intensive, (3) high memory (RAM) demanding, and (4) balanced resource needs. The exact SPs' resource requirements are randomly chosen between the range reported in Table III. Each location is endowed with wireless communication resources and an edge computing facility with CPUs and RAM. All locations are also connected to three cloud servers that provide additional computing power, including CPUs and RAM. We neglect the energy cost of the RAM against radio and CPU usage.

At each location l , radio resources are shared between connecting the MEC facility and the cloud. The total radio resource usage at location l is therefore the sum of all radio resources allocated across all services s and cloud nodes c ,

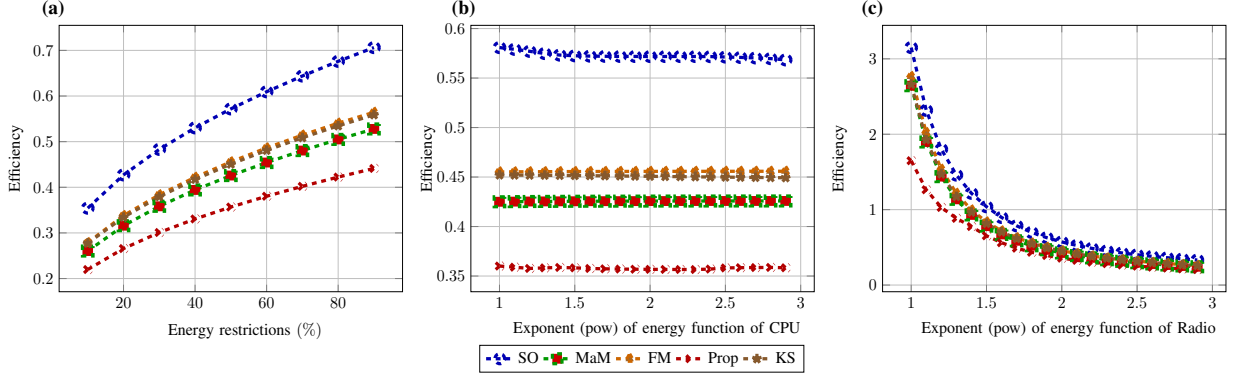


Fig. 2: Sensitivity of Efficiency to various factors: (a) restrictions on energy consumption (b) the CPU energy function, (c) the Radio energy function

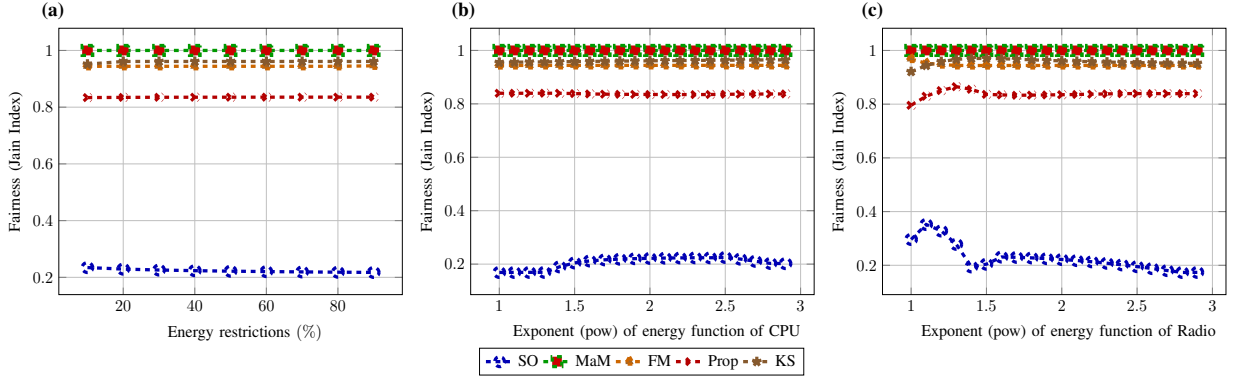


Fig. 3: Sensitivity of Fairness to various factors: (a) restrictions on energy consumption (b) the CPU energy function, (c) the Radio energy function

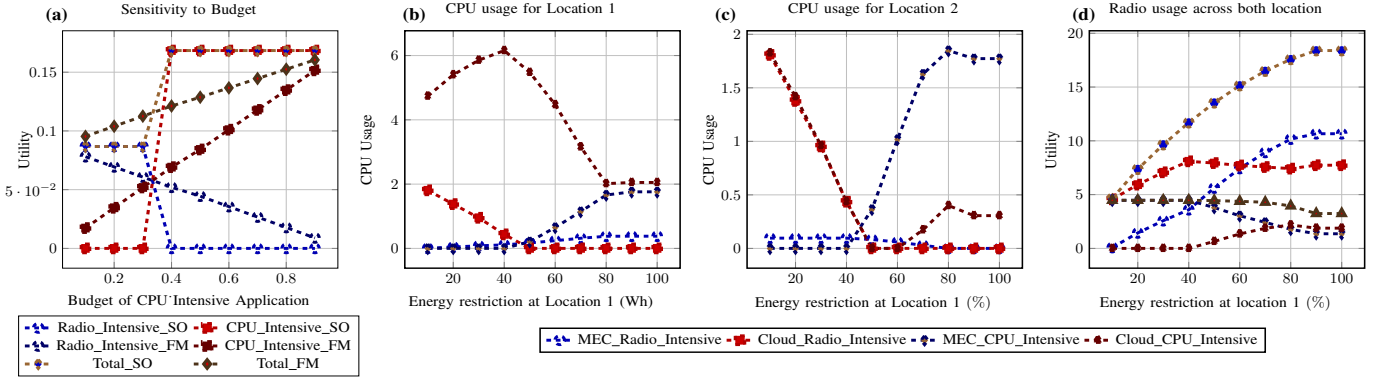


Fig. 4: Variation in the usage of: (a) radio, (b) CPU, and the utility derived from both MEC and cloud facilities vs. local energy consumption constraints.

TABLE III: The base demand vector of service classes

User Applications	VCPU	RAM (GB)	NB (Mbps)
BW-Intensive	2-4	8-12	300-492
CPU-Intensive	30-36	6-8	50-70
RAM-Intensive	2-4	28-32	50-70
Balanced	2-4	3.5-4	50-70

given by: $\hat{\mathbf{x}}_{\text{RAN},l} = \sum_s \sum_c x_{\text{RAN},lc}^s$. While, the total CPU and RAM demands at the MEC facility located at l are calculated by summing the demands from all services s at the MEC node (denoted by $c = 0$) as: $\hat{\mathbf{x}}_{\text{CPU},l} = \sum_s x_{\text{CPU},l0}^s$ and $\hat{\mathbf{x}}_{\text{RAM},l} = \sum_s x_{\text{RAM},l0}^s$. Similarly, let $\tilde{\mathbf{x}}_{\text{CPU},c} = \sum_s \sum_l x_{\text{CPU},lc}^s$ and $\tilde{\mathbf{x}}_{\text{CPU},c} =$

$\sum_s \sum_l x_{\text{RAM},lc}^s$ denote the total demand for resource CPU and RAM at cloud facility c . The total energy consumption due to RAN and CPU at the each location l is given as

$$\hat{e}_l(\hat{\mathbf{x}}_l) = \hat{e}_{\text{RAN},l}(\hat{x}_{\text{RAN},l}) + \hat{e}_{\text{CPU},l}(\hat{x}_{\text{CPU},l}) \quad \forall l \in \mathcal{L},$$

and the total energy consumption in the clouds is given as

$$\tilde{e}_c(\tilde{\mathbf{x}}_c) = \tilde{e}_{\text{CPU},c}(\tilde{x}_{\text{CPU},c}) \quad \forall c \in \mathcal{C},$$

where we consider that the energy consumption for each RAN and CPU resource follows a power function: $e_k(\bar{x}_k) = a_k(\bar{x}_k)^{\beta_k}$, where $\beta_k \geq 1$ and $a_k \geq 0$. This formulation helps

to analyze different trade-offs in energy consumption. Also, in case of CO₂ constraints, the coefficient a_k is an abstract yet realistic way of representing different energy mix at different locations. In this paper, we do not focus on the exact form of the energy functions. Instead, we adopt a general parametric energy model to encompass a wide range of computing and communication power behaviors reported in the literature, from linear to cubic [35]–[38].

The parameters (a_k) and (β_k) are deployment-specific and characterize the energy profile and carbon footprint of resource (k) . In practice, they can be estimated from measurements of resource utilization and power consumption using standard curve-fitting techniques, such as nonlinear least-squares regression. The carbon-intensity component can be updated dynamically according to the local electricity mix. In practical deployments, such information can be obtained from near-real-time carbon-intensity tracking services provided by electricity-market operators and carbon-aware platforms (e.g., RTE Eco2Mix³ and Electricity Maps⁴). For the power-law model ($e(x) = ax^\beta$), the parameters can be obtained by fitting measured utilization-power pairs $((x_i, e_i))$, either directly through nonlinear regression or via linear regression in the log-log domain. Similar measurement-based calibration approaches are commonly used for cellular base stations, communication infrastructures, and data-center servers [39]–[42]. The proposed framework does not rely on a particular choice of (a_k) and (β_k) and remains applicable to any convex energy-consumption model. Our methodology only requires that the energy functions be convex; if empirical measurements result in non-convex models, standard convex approximation techniques, such as piecewise-linear approximations or successive convex approximation, can be applied without affecting the distributed nature of the proposed algorithm. Using this framework, we systematically explore the impact of the parameters β_k and a_k on the overall trade-off between system utility and costs (e.g., energy and carbon footprint).

We impose energy constraints at two levels: (i) overall energy consumption across all resources and (ii) local energy limits for radio resources and the MEC facility at each location. We apply the proposed resource allocation method by solving (12), and compare the resulting market equilibrium (ME) allocation with the social optimum (SO), which maximizes the total weighted service provided by SPs under energy and resource capacity constraints.

We compare the proposed Fisher Market (FM)–based resource allocation scheme with several baseline allocation schemes, namely the

- 1) *Social Optimum (SO)* The social optimum maximizes the weighted aggregate system utility

$$\max_{\mathbf{x} \in \mathcal{X}} \sum_{i \in \mathcal{N}} U_i(\mathbf{x}). \quad (17)$$

- 2) *Max–Min Fairness (MaM)*: The max-min fair allocation aims to maximize the utility of the worst-off user:

$$\max_{\mathbf{x} \in \mathcal{X}} \min_{i \in \mathcal{N}} U_i(\mathbf{x}). \quad (18)$$

This objective provides strong fairness guarantees but often results in reduced overall efficiency.

- 3) *Kalai–Smorodinsky (KS) Bargaining Solution*: Let $U_i^{\max} = \max_{\mathbf{x} \in \mathcal{X}} U_i(\mathbf{x})$ denote the maximum achievable utility of user i , and let d_i be the disagreement utility. The Kalai–Smorodinsky solution is defined as:

$$\begin{aligned} & \max_{\mathbf{x} \in \mathcal{X}, \lambda} \lambda \\ & \text{s.t. } \frac{U_i(\mathbf{x}) - d_i}{U_i^{\max} - d_i} \geq \lambda, \quad \forall i \in \mathcal{N}. \end{aligned} \quad (19)$$

- 4) *Proportional allocation (PROP)*: In general, proportional allocation mechanisms distribute resources in proportion to users’ budgets. However, in our setting, resource usage is constrained by energy consumption limits, which prevents allocating each resource at full capacity purely based on budgets. Therefore, we first compute the resource usage profile that maximizes the aggregate network utility under the energy constraints. The resulting resources are then allocated among users in proportion to their budgets.

In this work our one of the goals is to achieve a favorable trade-off between efficiency and fairness. Accordingly, we evaluate all schemes along these two dimensions. Efficiency is measured by the total system utility achieved under each scheme. Fairness, on the other hand, is quantified using the Jain’s fairness index computed over SPs’ achieved utilities. This comparison allows us to assess how well the proposed FM-based approach balances system efficiency against fairness relative to existing allocation mechanisms.

Figs. 2 illustrate the sensitivity of system efficiency to variations in cost functions and energy constraints across all allocation schemes. In Fig. 2(a), we examine the effect of relaxing the energy constraint at location 1 on the performance of different schemes. As the energy restriction is eased, the overall system efficiency improves. Notably, the proposed FM-based allocation mechanism consistently outperforms both the Max-Min (MaM) and Proportional (PROP) allocation schemes. Figs. 2(b) and (c), along with Figs. 3(b) and (c), present results obtained by varying the exponent of the cost functions, β_k , from 1 to 3. We observe that increasing the exponent of the CPU energy function leads to a decrease in overall utility; however, the system is more sensitive to changes in the exponent of the Radio energy function. Across all scenarios, the proposed FM-based method achieves higher efficiency compared to MaM and PROP, performs on par with the KS scheme, and reaches over 80% of the efficiency of the socially optimal allocation.

In terms of fairness, Fig. 3 illustrates the effect of (a) varying the energy restriction, (b) changing the exponent of the CPU energy cost function, and (c) modifying the exponent of the radio energy cost function. We observe that the proposed allocation scheme achieves a high level of fairness, remaining close to the optimal fairness provided by the MaM scheme and nearly equivalent to the KS scheme across all considered parameter variations. Overall, the results in Figs. 2 and 3 demonstrate that the proposed FM-based allocation mechanism achieves a favorable balance between efficiency

³<https://www.rte-france.com>

⁴<https://app.electricitymaps.com>

and fairness. In particular, across all considered scenarios with Leontief utilities, the proposed scheme increases Jain's fairness index to approximately 60%–75%, while incurring only a 12%–22% reduction in aggregate utility. These findings indicate that substantial improvements in proportional fairness can be achieved at a relatively modest cost in overall system efficiency, highlighting the effectiveness of the proposed market-based allocation framework.

To simplify the analysis, in the remaining simulations shown in Figs. 4, we focus on two SPs: the first one offering a radio-intensive service, and the other a CPU-intensive service. In Fig. 4(a), we study how resource allocation is influenced by the SPs' budgets. Specifically, we vary the budget of the radio-intensive SP from 0.1 to 0.9 and observe the resulting utilities for both SPs under two allocation schemes. The results show that under the SO scheme, the SP with a lower budget receives significantly less utility. In contrast, the proposed ME-based approach achieves better fairness, is less sensitive to budget variations, and strikes a more balanced trade-off between cost, utility, and fairness.

Fig. 4 (b), (c), and (d) show how local energy constraints affect resource allocation and SPs utilities. For these simulations, we vary the energy limit at location 1 from 10 to 90 units, while keeping it fixed at other locations. Fig. 4(b) shows how CPU usage at MEC location 1 and the cloud facilities changes under different restriction levels. For the CPU-Intensive case, as restrictions are gradually relaxed, cloud usage initially increases since more radio resources become available for offloading. However, with further relaxation, MEC usage starts to increase, reducing dependency on the cloud. In contrast, for the Radio-Intensive case, the cloud is initially used more intensively. However, as restrictions ease, cloud usage decreases slightly while MEC usage increases, though not as significantly as in the CPU-Intensive case. Fig. 4(b) shows that even though the restrictions are applied only at location 1, they also impact CPU usage at location 2. This happens because both locations share two common cloud facilities and are subject to the same global energy constraints. In particular, we observe that as energy constraints are relaxed, both SPs reduce their cloud usage. To compensate, the cloud-intensive SP increases its use of MEC resources. The trends in Fig. 4(b) and (c) are further supported by Fig. 4(d), which shows how radio resource usage changes with varying restrictions. As restrictions at location 1 are relaxed, CPU-intensive SPs increasingly use radio to offload to the cloud, and similarly, radio-intensive SPs offload more to the MEC. In contrast, the use of MEC's CPU by CPU-intensive SPs decreases.

In addition to the Leontief utility model used in the baseline evaluation, we consider two alternative CCH utility functions to illustrate the applicability of the proposed framework beyond perfect-complementarity resource demands. The first utility function is given by

$$u_{lc}^{s,(A)}(\mathbf{x}_{lc}^s) = x_{\text{RAN},lc}^s + \min \left\{ \frac{x_{\text{CPU},lc}^s}{d_{\text{CPU},lc}^s}, \frac{x_{\text{RAM},lc}^s}{d_{\text{RAM},lc}^s} \right\}, \quad (20)$$

which introduces partial substitutability between radio and

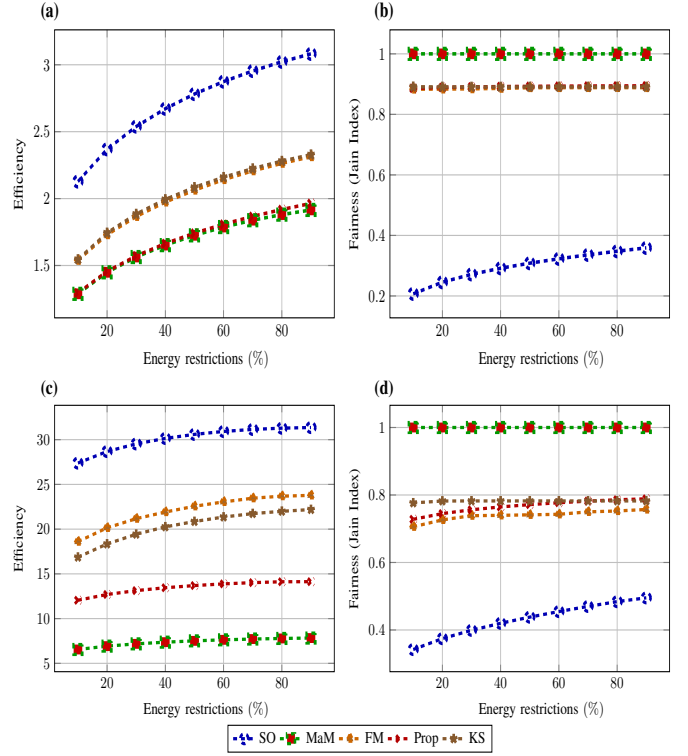


Fig. 5: Impact of Energy Consumption Constraints on Efficiency (a–c) and Fairness (d–f) for Utility Functions A and CD

computing resources. The second utility function is

$$u_{lc}^{s,(CD)}(\mathbf{x}_{lc}^s) = x_{\text{RAN},lc}^s \min \left\{ \frac{x_{\text{CPU},lc}^s}{d_{\text{CPU},lc}^s}, \frac{x_{\text{RAM},lc}^s}{d_{\text{RAM},lc}^s} \right\}, \quad (21)$$

which corresponds to a nested Cobb–Douglas–Leontief utility where radio resources interact multiplicatively with a computing component while CPU and RAM remain complementary. Both utility functions belong to the CCH class considered in this paper. Utility (20) is the sum of two degree-one homogeneous concave functions (a linear term and a Leontief term) and is therefore itself degree-one homogeneous and concave. Utility (21) is a nested Cobb–Douglas–Leontief utility; since the Cobb–Douglas composition preserves homogeneity and concavity for positive exponents and the inner Leontief component is homogeneous of degree one, this utility also satisfies the CCH assumptions. Consequently, all theoretical results developed in this paper apply directly to these utility specifications.

Fig 5 illustrates the impact of energy consumption constraints on efficiency (Figs. 5(a)–(c)) and fairness (Figs. 5(d)–(f)) for the utility functions in (20) and (21). For utility function (20), the proposed allocation mechanism achieves a fairness improvement of approximately 26%–53%, while incurring an efficiency reduction of 16%–36%. For utility function (21), the proposed mechanism yields an even larger fairness improvement of 50%–70%, at the cost of a relatively moderate efficiency loss of 24%–28%. The results demonstrate that the proposed framework can substantially improve fairness

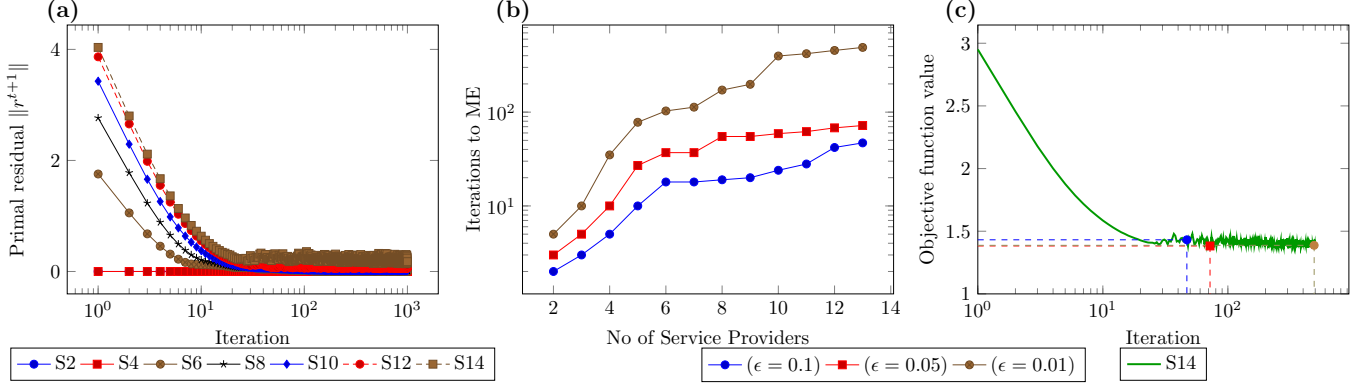


Fig. 6: Convergence behavior of ADMM under varying no of service providers

while maintaining a favorable efficiency–fairness trade-off under different utility specifications. Compared with the pure Leontief case, the additive and Cobb–Douglas utility models introduce increasing degrees of substitutability between radio and computing resources. As a result, resource bottlenecks become less severe and all allocation schemes achieve higher utility values. Nevertheless, the proposed market-based allocation consistently achieves the highest social welfare and resource utilization across all utility models.

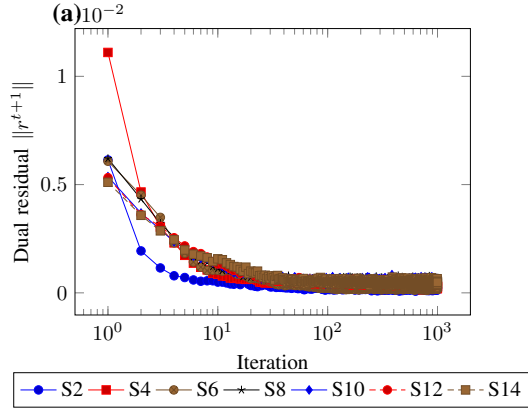


Fig. 7: Convergence behavior of the ADMM dual residual (price updates) under varying numbers of service providers.

Next, to demonstrate the practical viability and scalability of our proposed scheme, we implement the decentralized resource allocation method based on ADMM introduced in Section V. We evaluate the performance on a network with five locations and three cloud providers, considering varying numbers of service providers. Each service provider's base-line demand is randomly sampled from the range specified in Table III. Each experiment is repeated over 100 independent simulation instances, and the reported convergence performance corresponds to the average convergence behavior across these realizations. This provides statistically meaningful convergence benchmarks and demonstrates the scalability and robustness of the proposed algorithm. Fig. 6(a) illustrates the convergence behavior of the ADMM-based allocation method for a system with 2 to 14 SPs. Fig. 6(b) reports the minimum

number of iterations required to achieve convergence for two stopping tolerances, namely $\epsilon = 0.1$, $\epsilon = 0.05$ and $\epsilon = 0.01$, under different numbers of SPs. As expected, the smaller tolerance requires more iterations before convergence. However, Fig. 6(c) shows that the objective value stabilizes much earlier than the stopping criterion is satisfied, indicating that only marginal improvements are obtained by enforcing a tighter tolerance. Fig. 7 shows the convergence behavior of the ADMM dual residual, representing the evolution of the price updates, for different numbers of service providers. Moreover, the proposed algorithm employs an adaptive penalty parameter ρ , which is updated based on the primal and dual residuals. By automatically balancing primal and dual convergence, the adaptive penalty accelerates convergence and avoids the slow progress that may occur with an improperly chosen fixed ρ . Therefore, in practical deployments, a moderate stopping tolerance can achieve a favorable balance between computational latency and solution quality.

A. AI-based classification as a service

We now show the performance of the proposed resource allocation framework for a specific service, namely image classification based on pre-trained AI models that are available at edge and cloud facilities. The goal is to explore energy, accuracy and latency trade-offs.

TABLE IV: Model and JPEG Compression Statistics

(a) Model Computational Requirements		(b) JPEG Compression Levels	
Model	FLOPs	JPEG Quality (%)	Size (Bytes)
MobileNet V3	1.2×10^8	JPEG-20	1.8694×10^4
MobileNet V2	6.0×10^8	JPEG-40	2.3970×10^4
ResNet-18	3.6×10^9	JPEG-60	2.8833×10^4
ResNet-50	8.18×10^9	JPEG-80	3.2758×10^4
ResNet-101	1.56×10^{10}	JPEG-95	5.4397×10^4
ViT-B/16	3.36×10^{10}		

We define the *operation delay* for each service provider as the minimum time required to complete a task, including both *transmission* (denoted by D_{tx}) and *processing* (denoted by D_{proc}) delays, depending on whether the task is executed

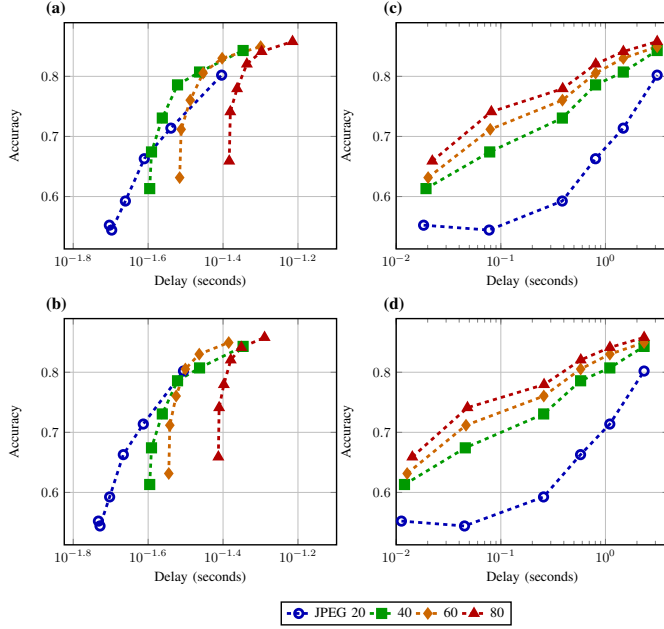


Fig. 8: Accuracy–delay trade-off for different JPEG compression levels under two energy budgets: 40% for SP1 (a) and SP2 (c), and 80% for SP1 (b) and SP2 (d).

at a Mobile Edge Computing (MEC) node or offloaded to a remote cloud facility. The total delay is,

$$D_{\text{total}} = \underbrace{\frac{b\dot{M}}{\tau}}_{D_{\text{tx}}} + \underbrace{\frac{\kappa\dot{M}}{\varphi}}_{D_{\text{proc}}} + \underbrace{\delta_{\text{cloud}}}_{\text{if cloud is selected}}, \quad (22)$$

where M is the number of mobile users served by the SP⁵, b is the input data size (in bits), τ is the transmission rate (in bits/s), κ is the computational workload (in FLOPs), φ is the allocated computing capacity (in FLOPs/s), δ_{cloud} is the additional latency incurred when the cloud is selected. Obviously, τ and φ depend on the allocated radio and computing resources, respectively (via the proposed method), while b and κ depend on the quality of the transmitted data (e.g., compression scheme) and the AI model used for inference, respectively. Denoting by W the allocated bandwidth (in Hz), by M the modulation scheme (e.g., QPSK, 16-QAM, etc.), and by R_c the coding rate, τ reads as

$$\tau = R_c \cdot W \cdot \log_2(M). \quad (23)$$

The exploited models⁶ with the corresponding computational demand and the JPEG compression schemes with the corresponding data size are reported in Table IV. Pre-trained models on ImageNet-1K are fine tuned on CIFAR-100⁷. We consider energy consumption function as quadratic function both for CPU and Radio resources.

⁵For simplicity, we assume that the SP obtained resources (wireless and computing) are equally shared among all users. More complex dynamic resource allocation schemes among users as in [43], [44] will be considered in future investigations.

⁶<https://docs.pytorch.org/vision/main/models.html>

⁷<https://docs.pytorch.org/vision/main/datasets.html>

We consider two service providers (SPs) with distinct resource requirements. We assume that data quality depends on the selected JPEG quality. The first service provider (SP1) uses JPEG-20 compressed images and a vision transformer model (namely, vit_b_16) for inference, resulting in higher computational demand. In contrast, the second service provider (SP2) uses JPEG-95 images and the lightweight MobileNet V3 model, which reduces computation but significantly increases the input data size. As a result, SP1 is characterized as a CPU-intensive application, while SP2 is primarily bandwidth-intensive, due to the larger input data size that must be transmitted for processing. We assume that both SPs allocate resources proportionally to the number of users they serve, and that each is capable of supporting multiple users concurrently. Specifically, SP1 serves 15 users, while SP2 serves 20 users. We use our FM based resource allocation scheme, and evaluate the performance under various settings.

To evaluate the impact of energy constraints, we systematically vary the energy budget at each location. We consider that the global energy consumption constraints are 2000 (Wh) units. Additionally, we explore the effect of energy sensitivity by varying the coefficients of the quadratic energy functions for both CPU and radio resources. These evaluations consider multiple inference models operating under the same resource allocation.

We evaluate the delay performance of the proposed resource allocation scheme under various operational scenarios. Fig. 8(a)-(b) and Fig. 8(c)-(d) depict the trade-off between inference delay and accuracy under varying energy constraints 40% and 80% of global energy constraints at a fixed location for service providers SP1 and SP2, respectively. We observe that tightening the energy constraint at the location has a relatively minor impact on the delay experienced by SP1 compared to SP2 across most models and JPEG quality levels. The effect becomes noticeable for more advanced models and higher-quality JPEG files, where stricter energy limitations lead to increased inference delay.

In contrast, SP2 experiences increased delay when the energy constraint is tightened (e.g., to 200 (Wh) units), and this delay is higher even in more advanced models and higher-quality JPEG files. This is because radio resources are only available at the location, and both MEC and cloud services rely on wireless connectivity. At the location, the available energy is primarily used to power the radio interface, and any remaining energy is allocated to applications with low CPU demands (SP2). When energy is limited, the location cannot support local computation, forcing SP2 to rely on the cloud, which results in higher delay. For SP1, which is primarily served by the cloud, the impact of energy restrictions on delay is comparatively smaller, though high-quality JPEG files and complex models still exhibit some increase in latency.

Fig. 9(a) and Fig. 10(a) show the effect of varying the coefficient in the CPU energy consumption function. This variation has little impact on the overall resource allocation. In contrast, changing the coefficient in the radio energy consumption function significantly affects the delay–accuracy trade-off for SP2, as shown in Fig. 10(b): as the coefficient increases, the delay also increases.

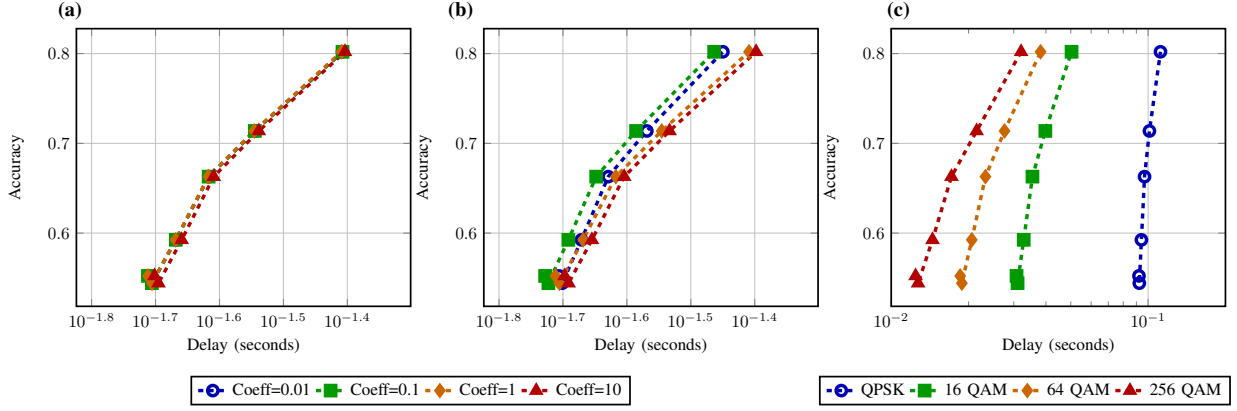


Fig. 9: Accuracy vs. Delay for SP1 under varying conditions: (a) varying CPU energy cost coefficients, (b) varying radio energy cost coefficients and (c) different modulation schemes

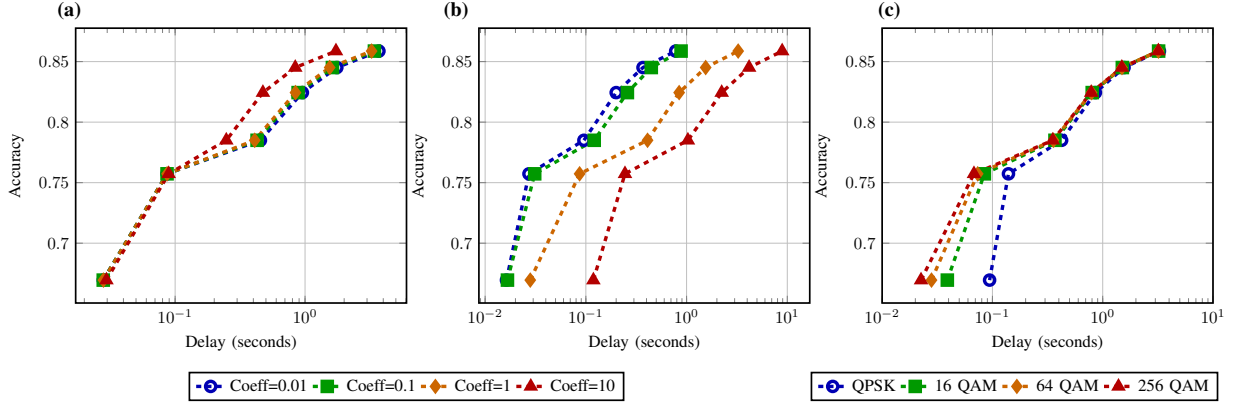


Fig. 10: Accuracy vs. Delay for SP2 under varying conditions: (a) varying CPU energy cost coefficients, (b) varying radio energy cost coefficients, and (c) different modulation schemes

In Fig. 9(c) and Fig. 10(c), we analyze a scenario where service providers use different modulation schemes (cf. (23)): QPSK, 16-QAM, 64-QAM, and 256-QAM. Each scheme uses a specific coding rate 0.5, 0.75, 0.83, and 0.93 respectively. If all schemes use the same bandwidth, they provide different data rates. Specifically, the effective spectral efficiencies for these coding schemes are 1, 3, 4.98, and 7.44 bits/s/Hz for QPSK, 16-QAM, 64-QAM, and 256-QAM, respectively. Our results show that for SP1, changes in the modulation scheme have a more pronounced impact on the overall delay since its service is data-intensive and predominantly affected by transmission delays. In contrast, for SP2, the effect of modulation schemes is more significant for lightweight models, whereas for larger models, the delay is primarily dominated by computation time rather than transmission latency.

VII. CONCLUSIONS AND FUTURE WORK

We proposed a novel resource allocation and pricing mechanism based on an extended Fisher market model that accounts for heterogeneous resources including radio, computing and external costs such as energy consumption and environmental impact. These external costs vary spatially across different locations, for example due to differing regulations or energy sources. By incorporating Pigouvian taxes, our approach ensures that resource constraints support sustainability

while maximizing both Nash welfare and proportional fairness among agents. We show that the market equilibrium can be computed using convex optimization. Furthermore, we propose a decentralized resource allocation scheme that demonstrates the method's low complexity, practicality, and effectiveness.

Main limitations for future investigations. the proposed framework relies on convexity assumptions for utility and energy-consumption functions, which are necessary to guarantee tractability and convergence to a ME. In practical network deployments, however, resource-performance relationships and energy models may be non-convex, time-varying, or difficult to estimate accurately. One way to extend this aspect is to consider realistic data and fit convex functions to apply the methodology.

Further, while the FM formulation enables decentralized decision-making, the approach still requires a market operator or coordinating entity to define budgets, enforce energy constraints, and update prices or taxes associated with resource usage. Finally, the numerical evaluation demonstrates the effectiveness of the proposed mechanism under simulated and application-specific scenarios, but further validation using real network traces, operational cloud-edge platforms, and dynamic traffic conditions would be needed to fully assess its practical applicability.

APPENDIX

Consider Lagrangian for optimization program (12)

$$\mathcal{L}(\mathbf{z}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = \sum_{n \in \mathcal{N}} B^n \log(\phi^n(\mathbf{z}^n)) - \sum_{i \in \mathcal{I}} \lambda_i (e_i(\bar{\mathbf{z}}) - E_i) - \sum_{k \in \mathcal{K}} \gamma_k \left(\sum_n z_k^n - C_k \right) + \nu_{nk} z_k^n.$$

Let us apply the Karush–Kuhn–Tucker (KKT) conditions [45].

Stationarity Conditions: $\forall k \in \mathcal{K}, \forall n \in \mathcal{N}$, the derivative of the Lagrangian with respect to z_k^n is zero at the optimum:

$$\left[\frac{B^n \nabla_{nk} \phi^n(\mathbf{z}^n)}{\phi^n(\mathbf{z}^n)} \right]_{\mathbf{z}=\mathbf{z}^*} - \sum_{i \in \mathcal{I}} \lambda_i^* [\nabla_k e_i(\bar{\mathbf{z}})]_{\mathbf{z}=\mathbf{z}^*} - \gamma_k^* + \nu_{nk}^* = 0.$$

If we replace $\sum_{i \in \mathcal{I}} \lambda_i^* [\nabla_k e_i(\bar{\mathbf{z}})]_{\mathbf{z}=\mathbf{z}^*} - \gamma_k^* = p_k^*$, $\forall k \in \mathcal{K}$ as defined in (C2.1) we get

$$\left[\frac{B^n \nabla_{nk} \phi^n(\mathbf{z}^n)}{\phi^n(\mathbf{z}^n)} \right]_{\mathbf{z}=\mathbf{z}^*} - p_k^* + \nu_{nk}^* = 0, \quad \forall n \in \mathcal{N}, \forall k \in \mathcal{K}. \quad (24)$$

Primal Feasibility: $\forall n \in \mathcal{N}, \forall k \in \mathcal{K}, \forall i \in \mathcal{I}$,

$$z_k^{n*} \geq 0, \quad e_i(\bar{\mathbf{z}}^*) \leq E_i, \quad \sum_n z_k^{n*} \leq C_k,$$

Dual Feasibility: $\forall n \in \mathcal{N}, \forall k \in \mathcal{K}, \forall i \in \mathcal{I}$,

$$\lambda_i^* \geq 0, \quad \gamma_k^* \geq 0, \quad \nu_{nk}^* \geq 0,$$

Complementary Slackness: $\forall n \in \mathcal{N}, \forall k \in \mathcal{K}, \forall i \in \mathcal{I}$,

$$\lambda_i^* (e_i(\bar{\mathbf{z}}^*) - E_i) = 0, \\ \gamma_k^* \left(\sum_{n \in \mathcal{N}} z_k^{n*} - C_k \right) = 0, \quad \nu_{nk}^* z_k^{n*} = 0$$

Now consider the Lagrangian of the decision problem (11) of each agent $n \in \mathcal{N}$

$$\mathcal{L}_n(\mathbf{z}^n, \lambda_n, \mu_n) = \phi^n(\mathbf{z}^n) - \mu_n \left(\sum_k p_k^* z_k^n - B^n \right) - \sum_k \xi_{nk} z_k^n \quad (25)$$

Consider the KKT conditions at the best response

Stationarity:

$$[\nabla_k^n \phi^n(\mathbf{z}^n)]_{\mathbf{z}^n=\mathbf{z}^{n*}} - \mu_n^* p_k + \xi_{nk}^* = 0, \quad \forall k. \quad (26)$$

Primal feasibility:

$$\sum_k p_k z_k^{n*} \leq B^n, \quad z_k^{n*} \geq 0, \quad \forall k.$$

Dual feasibility: $\mu_n^* \geq 0, \quad \xi_{nk}^* \geq 0, \quad \forall k.$

Complementary slackness:

$$\mu_n^* \left(\sum_k p_k z_k^{n*} - B^n \right) = 0, \quad \xi_{nk}^* z_k^{n*} = 0, \quad \forall k.$$

Consider the Stationarity condition (26), multiplying both sides by z_k^{n*} and summing over all $k \in \mathcal{K}$, we have

$$[\nabla_{nk} \phi^n(\mathbf{z}^n)]_{\mathbf{z}^n=\mathbf{z}^{n*}} z_k^{n*} - \mu_n^* p_k z_k^{n*} + \xi_{nk}^* z_k^{n*} = 0, \quad \forall k. \quad (27)$$

$$\sum_k [\nabla_{nk} \phi^n(\mathbf{z}^n)]_{\mathbf{z}^n=\mathbf{z}^{n*}} z_k^{n*} - \mu_n^* B^n = 0, \quad \forall k. \quad (28)$$

If the utility function ϕ^n of each agent $n \in \mathcal{N}$ is homogeneous of degree 1, then by Euler's homogeneous function

theorem, we have: $\sum_k \nabla_{nk} \phi^n(\mathbf{z}^n) z_k^n = \phi^n(\mathbf{z}^n)$, replacing $\mu^n = \frac{\phi^n(\mathbf{z}^n)}{B^n}$ in (26)

$$\left[\frac{B^n \nabla_{nk} \phi^n(\mathbf{z}^n)}{\phi^n(\mathbf{z}^n)} \right]_{\mathbf{z}=\mathbf{z}^*} - p_k + \xi_{nk}^* = 0, \quad \forall n, k. \quad (29)$$

where $\xi_{nk}^* = \xi_{nk}^*/\mu_n^*$. From (24) and (29), when $\mathbf{p} = \mathbf{p}^*$ the KKT conditions at the optimum of problem (12) and the set of KKT conditions at the best response of the agents' decision problem (11) are equal, thus establishing the claim.

EQUIVALENCE OF PRIMAL UPDATES IN ADMM FORMULATION

We consider the primal update for agent s in the ADMM setup. The standard primal update derived from the augmented Lagrangian is given by:

$$\mathbf{x}^{s,(t+1)} = \arg \max_{\mathbf{x}^s \geq 0} \left\{ B^s \log \left(\sum_{l,c} u_{lc}^s(\mathbf{x}_{lc}^s) \right) + \left\langle \hat{\mathbf{p}}^{s,(t)}, \hat{\mathbf{x}}^s - \hat{\mathbf{y}}^{(t)} \right\rangle + \left\langle \tilde{\mathbf{p}}^{s,(t)}, \tilde{\mathbf{x}}^s - \tilde{\mathbf{y}}^{(t)} \right\rangle - \frac{\rho}{2} \left\| \hat{\mathbf{x}}^s - \hat{\mathbf{y}}^{(t)} \right\|^2 - \frac{\rho}{2} \left\| \tilde{\mathbf{x}}^s - \tilde{\mathbf{y}}^{(t)} \right\|^2 \right\} \quad (30)$$

This is equivalent to:

$$\mathbf{x}^{s,(t+1)} = \arg \max_{\mathbf{x}^s \geq 0} \left\{ B^s \log \left(\sum_{l,c} u_{lc}^s(\mathbf{x}_{lc}^s) \right) - \frac{\rho}{2} \left\| \hat{\mathbf{x}}^s - \hat{\mathbf{y}}^{(t)} + \frac{1}{\rho} \hat{\mathbf{p}}^{s,(t)} \right\|^2 - \frac{\rho}{2} \left\| \tilde{\mathbf{x}}^s - \tilde{\mathbf{y}}^{(t)} + \frac{1}{\rho} \tilde{\mathbf{p}}^{s,(t)} \right\|^2 + \text{const} \right\} \quad (31)$$

Now consider the consensus form of ADMM, where all agents coordinate via global averages and a common dual variable. Next, let

$$\hat{\mathbf{x}}_{\text{avg}}^{(t)} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \hat{\mathbf{x}}^{s,(t)}, \quad \hat{\mathbf{y}}_{\text{avg}}^{(t)} = \frac{1}{|\mathcal{S}|} \hat{\mathbf{y}}^{(t)},$$

and similarly for the $\tilde{\cdot}$ variables. The optimal solution of the consensus-variable update satisfies. Thus, the primal update becomes:

$$\mathbf{x}^{s,(t+1)} = \arg \max_{\mathbf{x}^s \geq 0} \left\{ B^s \log \left(\sum_l \sum_c u_{lc}^s(\mathbf{x}_{lc}^s) \right) - \frac{\rho}{2} \left\| \hat{\mathbf{x}}^s - \hat{\mathbf{x}}^{s,(t)} + \hat{\mathbf{x}}_{\text{avg}}^{(t)} - \hat{\mathbf{y}}_{\text{avg}}^{(t)} + \frac{1}{\rho} \hat{\mathbf{p}}^{(t)} \right\|_2^2 - \frac{\rho}{2} \left\| \tilde{\mathbf{x}}^s - \tilde{\mathbf{x}}^{s,(t)} + \tilde{\mathbf{x}}_{\text{avg}}^{(t)} - \tilde{\mathbf{y}}_{\text{avg}}^{(t)} + \frac{1}{\rho} \tilde{\mathbf{p}}^{(t)} \right\|_2^2 \right\} \quad (32)$$

1) Local Optimization

Given the broadcast variables

$$\left(\hat{\mathbf{x}}_{\text{avg}}^{(t)}, \tilde{\mathbf{x}}_{\text{avg}}^{(t)}, \hat{\mathbf{y}}_{\text{avg}}^{(t)}, \tilde{\mathbf{y}}_{\text{avg}}^{(t)}, \hat{\mathbf{p}}^{(t)}, \tilde{\mathbf{p}}^{(t)}, \rho^{(t)} \right)$$

each SP s solves the local optimization problem (32)

Since the utility function is concave and the quadratic penalty terms are strongly concave, the local subproblem is strictly concave and admits a unique optimum. In our implementation, each local optimization problem is solved using a convex optimization solver through CVXPY

2) Centralized Consensus Update

After receiving all local solutions, the central operator computes the average allocations:

$$\hat{\mathbf{x}}_{\text{avg}}^{(t+1)} = \frac{1}{S} \sum_{s=1}^S \hat{\mathbf{x}}_s^{(t+1)} \quad \tilde{\mathbf{x}}_{\text{avg}}^{(t+1)} = \frac{1}{S} \sum_{s=1}^S \tilde{\mathbf{x}}_s^{(t+1)}. \quad (33)$$

3) Global Projection STEP

The operator then computes a projection onto the feasible resource allocation set $\mathcal{C}$: Equivalently, the projection step solves the quadratic problem

$$(\hat{\mathbf{y}}^{(t+1)}, \tilde{\mathbf{y}}^{(t+1)}) = \underset{(\hat{\mathbf{y}}, \tilde{\mathbf{y}}) \in \mathcal{C}}{\text{argmin}} \left\{ \frac{S\rho}{2} \left\| \hat{\mathbf{y}} - \hat{\mathbf{x}}_{\text{avg}}^{(t+1)} - \frac{1}{\rho} \hat{\mathbf{p}}^{(t)} \right\|^2 + \frac{S\rho}{2} \left\| \tilde{\mathbf{y}} - \tilde{\mathbf{x}}_{\text{avg}}^{(t+1)} - \frac{1}{\rho} \tilde{\mathbf{p}}^{(t)} \right\|^2 \right\} \quad (34)$$

Since the objective function is concave and the quadratic term is strongly concave, it admits a unique optimum. In our implementation, projection is solved using a convex optimization solver through CVXPY. Now computing the average auxiliary variables we get

$$\hat{\mathbf{y}}_{\text{avg}}^{(t+1)} = \frac{1}{S} \hat{\mathbf{y}}^{(t+1)} \quad \tilde{\mathbf{y}}_{\text{avg}}^{(t+1)} = \frac{1}{S} \tilde{\mathbf{y}}^{(t+1)}. \quad (35)$$

4) Dual Variable Update

The dual variables associated with the consensus constraints are updated as

$$\hat{\mathbf{p}}^{(t+1)} = \hat{\mathbf{p}}^{(t)} + \rho^{(t)} \left(\hat{\mathbf{x}}_{\text{avg}}^{(t+1)} - \hat{\mathbf{y}}_{\text{avg}}^{(t+1)} \right), \quad (36)$$

$$\tilde{\mathbf{p}}^{(t+1)} = \tilde{\mathbf{p}}^{(t)} + \rho^{(t)} \left(\tilde{\mathbf{x}}_{\text{avg}}^{(t+1)} - \tilde{\mathbf{y}}_{\text{avg}}^{(t+1)} \right). \quad (37)$$

5) Residual Computation

The primal residual is defined as

$$\mathbf{r}^{(t+1)} = \left[\hat{\mathbf{x}}_{\text{avg}}^{(t+1)} - \hat{\mathbf{y}}_{\text{avg}}^{(t+1)}, \tilde{\mathbf{x}}_{\text{avg}}^{(t+1)} - \tilde{\mathbf{y}}_{\text{avg}}^{(t+1)} \right], \quad (38)$$

while the dual residual is

$$\mathbf{s}^{(t+1)} = \rho^{(t)} \left[\hat{\mathbf{y}}_{\text{avg}}^{(t+1)} - \hat{\mathbf{y}}_{\text{avg}}^{(t)}, \tilde{\mathbf{y}}_{\text{avg}}^{(t+1)} - \tilde{\mathbf{y}}_{\text{avg}}^{(t)} \right]. \quad (39)$$

Algorithm 2 Adaptive Penalty Update and Dual Scaling

```

1: if  $(t+1) \bmod T_\rho = 0$  then
2:   if  $\|\mathbf{r}^{(t+1)}\| > \varepsilon_r$  and  $\|\mathbf{s}^{(t+1)}\| > \varepsilon_s$  then
3:     if  $\|\mathbf{r}^{(t+1)}\| > \mu \|\mathbf{s}^{(t+1)}\|$  then
4:        $\rho^{(t+1)} \leftarrow \tau_{\text{inc}} \rho^{(t)}$ 
5:       Rescale dual variables:
6:        $\hat{\mathbf{p}}^{(t+1)} \leftarrow \hat{\mathbf{p}}^{(t+1)} \frac{\rho^{(t)}}{\rho^{(t+1)}}$  and  $\tilde{\mathbf{p}}^{(t+1)} \leftarrow \tilde{\mathbf{p}}^{(t+1)} \frac{\rho^{(t)}}{\rho^{(t+1)}}$ 
7:     else if  $\|\mathbf{s}^{(t+1)}\| > \mu \|\mathbf{r}^{(t+1)}\|$  then
8:        $\rho^{(t+1)} \leftarrow \rho^{(t)} / \tau_{\text{dec}}$ 
9:       Rescale dual variables:
10:       $\hat{\mathbf{p}}^{(t+1)} \leftarrow \hat{\mathbf{p}}^{(t+1)} \frac{\rho^{(t)}}{\rho^{(t+1)}}$  and  $\tilde{\mathbf{p}}^{(t+1)} \leftarrow \tilde{\mathbf{p}}^{(t+1)} \frac{\rho^{(t)}}{\rho^{(t+1)}}$ 
11:    else
12:       $\rho^{(t+1)} \leftarrow \rho^{(t)}$ 
13:    end if
14:  end if
15: end if

```

6) Adaptive Penalty Parameter Update

To balance the convergence of the primal and dual residuals, we employ an adaptive penalty update strategy based on the residual balancing heuristic. Every T_ρ iterations, the algorithm compares the norms of the primal and dual residuals, $\mathbf{r}^{(t+1)}$ and $\mathbf{s}^{(t+1)}$, respectively. The update is performed only when both residuals exceed their corresponding convergence tolerances,

$$\|\mathbf{r}^{(t+1)}\| > \varepsilon_r, \quad \|\mathbf{s}^{(t+1)}\| > \varepsilon_s,$$

thereby avoiding unnecessary modifications of the penalty parameter once the algorithm is close to convergence. The penalty parameter is updated according to

$$\rho^{(t+1)} = \begin{cases} \tau_{\text{inc}} \rho^{(t)}, & \text{if } \|\mathbf{r}^{(t+1)}\| > \mu \|\mathbf{s}^{(t+1)}\|, \\ \rho^{(t)} / \tau_{\text{dec}}, & \text{if } \|\mathbf{s}^{(t+1)}\| > \mu \|\mathbf{r}^{(t+1)}\|, \\ \rho^{(t)}, & \text{otherwise,} \end{cases} \quad (40)$$

where $\mu > 1$ is a balancing threshold, while $\tau_{\text{inc}} > 1$ and $\tau_{\text{dec}} > 1$ denote the multiplicative increase and decrease factors, respectively. Since the scaled dual variables explicitly depend on the penalty parameter, changing ρ without any adjustment would alter their effective values. Therefore, after updating ρ , the scaled dual variables are rescaled according to

$$\hat{\mathbf{p}}^{(t+1)} \leftarrow \hat{\mathbf{p}}^{(t+1)} \frac{\rho^{(t)}}{\rho^{(t+1)}}, \quad (41)$$

$$\tilde{\mathbf{p}}^{(t+1)} \leftarrow \tilde{\mathbf{p}}^{(t+1)} \frac{\rho^{(t)}}{\rho^{(t+1)}}. \quad (42)$$

This rescaling preserves the equivalence between the scaled and unscaled forms of ADMM and ensures that the dual variables remain consistent following the adaptive update of the penalty parameter.

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